ELARA: Reliability Gates Under Protocol Limits — A Cross-Benchmark Study of Drift-Signal Generalization in Multimodal Anomaly Fusion

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Abstract—**We report two complementary results on validation-derived drift gates in multimodal anomaly fusion. First, on a label-aligned benchmark with both classes present at training time, the Kolmogorov–Smirnov drift signal carries the entire all-domain coherent score-collapse robustness gain (+0.0506 `zero_attack`, +0.0319 `max_attack`) delivered by the reliability gate; removing the KS-drift component eliminates the gain. A learned-gate alternative does not match the heuristic's gain. Second, on the naturally paired MVTec 3D-AD benchmark under its canonical one-class anomaly-detection protocol (normal-only training, mixed test), every supervised fusion baseline – random forest, MLP, late-fusion ensemble, Tent, TTT, static attention, RGA – remains near chance ROC-AUC because none of them can learn from one-class training. The gate's clean delta is small there. We include held-out-category, PatchCore-style, and supervised-paired public MVTec variants to test whether the conclusion is an artefact of the first scorer or split. The combined finding is methodological: validation-derived drift gates are useful as diagnostics for coherent score-collapse stress, but they are not universal clean-performance winners. A validation-selected RGA+ head adds reliability-derived features and gradient-boosted fusion; on the public PatchCore supervised-paired variant it becomes the best ROC-AUC method among pre-specified heads (0.738 versus Tent 0.735), while an exploratory validation router reaches 0.740. On the held-out-category split, RGA+ gives only a marginal ROC-AUC gain over TTT (0.517 versus 0.516), and TTT remains slightly ahead on PR-AUC.**

The result is produced inside ELARA (Evidence-Layered Anomaly Reliability Architecture), a research prototype for score-level multimodal anomaly fusion. The mechanism under study is Reliability-Gated Attention (RGA), which couples masked cross-domain attention with a post-hoc reliability estimator (validation expected calibration error, test-time KS drift, prediction sharpness) and a conservative gate that preserves the static attention model unless mean reliability falls below a threshold. The system bundle includes a reproducible long-format fusion schema, strong score-level baselines (random forest, early-fusion MLP, late-fusion ensemble, Tent and TTT score adapters, static attention), missing-domain ablations, adversarial perturbation stress tests, calibration analysis, and counterfactual domain attribution.

The empirical study spans one external public paired benchmark family and one label-aligned stress benchmark. The primary benchmark family is naturally paired MVTec 3D-AD across eight categories (3,226 paired RGB/depth samples, 22.4% positive). The secondary benchmark, ELARA-Bench-LA, combines four real local datasets (Credit Card Fraud, UNSW-NB15, Online Shoppers Intention, labeled news text) with explicit label alignment. All benchmark configs honor disciplined train/validation/test boundaries so the fusion test fold is disjoint from the rows the per-domain scorers fit on. On MVTec 3D-AD under its canonical one-class anomaly-detection protocol, all supervised fusion baselines converge near chance ROC-AUC (random forest 0.500, static attention 0.542, RGA 0.561); the gate's delta to static is small and slightly positive ($+0.018$). On ELARA-Bench-LA the same gate improves all-domain `zero_attack` and `max_attack` ROC-AUC by $+0.0506$ and $+0.0319$ respectively. On the public MVTec PatchCore supervised-paired protocol, base RGA remains below Tent, but the validation-selected RGA+ boosted fusion reaches ROC-AUC 0.738, above Tent (0.735), TTT (0.724), and random forest (0.702); the auxiliary validation router reaches 0.740. Component ablation shows the KS-drift signal controls whether the gate fires; removing it eliminates the ELARA-Bench-LA gain. We report these benchmarks with confidence intervals and explicit mechanism isolation. The MVTec result is a protocol diagnostic, not a method critique: any supervised fusion benchmark that trains on one class only will see every supervised method collapse together, so the clean-fusion question is undefined there.

Index Terms—**anomaly detection, multimodal fusion, reliability, calibration, attention, score-level fusion, counterfactual explanation, fraud detection, cybersecurity**

I. Introduction

Operational anomaly detection is usually a system problem before it is a single-model problem. A high-risk case may appear weakly in transaction records, network telemetry, user behavior, text evidence, device signals, or document checks. Each domain expert may be accurate inside its own data distribution, but the final decision still depends on a fusion layer that decides how much to trust each signal. This trust problem becomes harder when domains are missing, score distributions drift, calibration degrades, or an adversary corrupts one or more sources.

ELARA is a research system for this setting. The system is not designed as a new fraud detector, network detector, or text detector in isolation. Instead, it treats anomaly detection as a verification pipeline: domain experts generate scores and compact evidence, those scores are converted into a unified fusion schema, and a reliability-aware fusion module produces a final risk estimate with a domain-level explanation. This architecture matches how many practical anomaly systems are deployed: detectors are built and maintained by different teams, and the fusion layer must remain robust as their data quality changes.

The paper treats ELARA as the system boundary and RGA as one mechanism inside that boundary. This matters because the contribution is not a new domain detector. It is a reliability-aware fusion layer with explicit input schema, stress tests, and limitations.

A. Research Question

The paper asks:

Can a multimodal anomaly-verification system preserve clean score-fusion accuracy while using calibration and drift evidence to reduce brittle behavior under domain failure?

This question is intentionally narrower than a claim of general anomaly detection superiority. The present codebase already supports domain experts, fusion schemas, attention fusion, reliability scoring, and robustness tests. The research target is the fusion and verification layer that combines those experts under imperfect operating conditions.

B. Thesis and Evidence Standard

The working thesis is that reliability-aware fusion is most useful when the operating condition changes. A strong paper must therefore show three things:

1) clean performance remains competitive with strong baselines,
2) reliability adaptation improves behavior under meaningful stress, and
3) the data and experimental protocol justify the scope of the claim.

The current results support the first two conditions for score-level fusion but do not yet fully satisfy the third condition for co-observed multimodal incidents. This limitation is explicit throughout the manuscript.

C. Contributions

We make four scoped contributions:

- A cross-benchmark scope result for validation-derived drift gates. The Kolmogorov–Smirnov drift signal carries the all-domain stress-test gain on the label-aligned secondary benchmark, while the naturally paired MVTec 3D-AD path shows that canonical one-class anomaly protocols do not provide a standard two-class supervised-fusion test bed.
- RGA couples masked cross-modal attention with a three-signal reliability estimator (validation ECE, test-time KS drift, score sharpness) and a conservative gate that preserves the static path unless mean reliability falls below τ.
- ELARA-Bench-LA is a reproducible the secondary benchmark real-domain score-fusion benchmark over Credit Card Fraud, UNSW-NB15, Online Shoppers Intention, and labeled news text. A naturally paired MVTec 3D-AD benchmark path covers eight categories with predefined-split discipline.
- ELARA packages the schema, baselines (random forest, early-fusion MLP, late-fusion ensemble, Tent, TTT, static attention, RGA), the validation-selected RGA+ boosted/router head, missing-domain ablation, drift curves, adversarial perturbation stress tests, calibration analysis, and counterfactual domain attribution with an explicit construct-validity boundary.

TABLE I
SYSTEM REQUIREMENTS FOR ELARA AND HOW THE CURRENT IMPLEMENTATION ADDRESSES THEM.

Requirement	Current implementation
Heterogeneous evidence	Fraud, cyber, behavior, and text domains are mapped to a shared score-level schema.
Missing-domain support	The attention model uses explicit masks; baselines receive missingness-aware fallbacks.
Reliability awareness	Domain weights combine calibration error, KS drift evidence, and score sharpness.
Strong baselines	Random forest, early fusion MLP, late fusion ensemble, static attention, and confidence weighting are included.
Explanation	Counterfactual domain attribution masks one domain at a time and measures risk change.
Reproducibility	Data generation, benchmark execution, asset generation, and PDF compilation are scriptable.

The evidence is strongest as a reliability-stress study over score-level fusion. It is not a general claim that RGA is always superior to static attention, nor as evidence that the label-aligned benchmark captures naturally co-observed multimodal incidents.

Figure 1 shows the system boundary. Table I summarizes the implementation requirements used to keep the paper aligned with the codebase.

II. Related Work

A. Anomaly Detection Systems

Deep anomaly detection has been studied across tabular, sequential, text, graph, and visual data. The classical formulation distinguishes statistical outliers from semantically anomalous patterns [26], [25]; density-based detectors [30], ensembles of isolation trees [31], and distance-based methods [32] remain widely used baselines. The deep-learning era added one-class SVMs and deep one-class classifiers [33], [34], [35], reconstruction-based detectors [37], [38], [39], [36], and the broader deep-AD surveys that summarise the field [3], [29], [27], [28]. For industrial vision specifically, recent surveys cover the explosion of unsupervised localisation methods [168], [169], [170]. ELARA follows the systems view emphasised by Pang et al. [3]: it does not assume a single representation can absorb every domain cleanly, but treats each domain expert as a separate source of evidence.

B. Multimodal and Score-Level Fusion

Multimodal learning surveys distinguish early fusion, late fusion, hybrid fusion, and interaction-based fusion [2], [172], [173], [171], [174], [175], [124]. Early fusion concatenates features before prediction; late fusion combines model outputs; stacking [166] learns a meta-model over domain predictions, which is the lineage of the RGA+ router we introduce later. Transformer attention [1] gives a natural mechanism for modelling interactions across available domains and is the basis for recent visual transformers [117], [123], [119], [122];

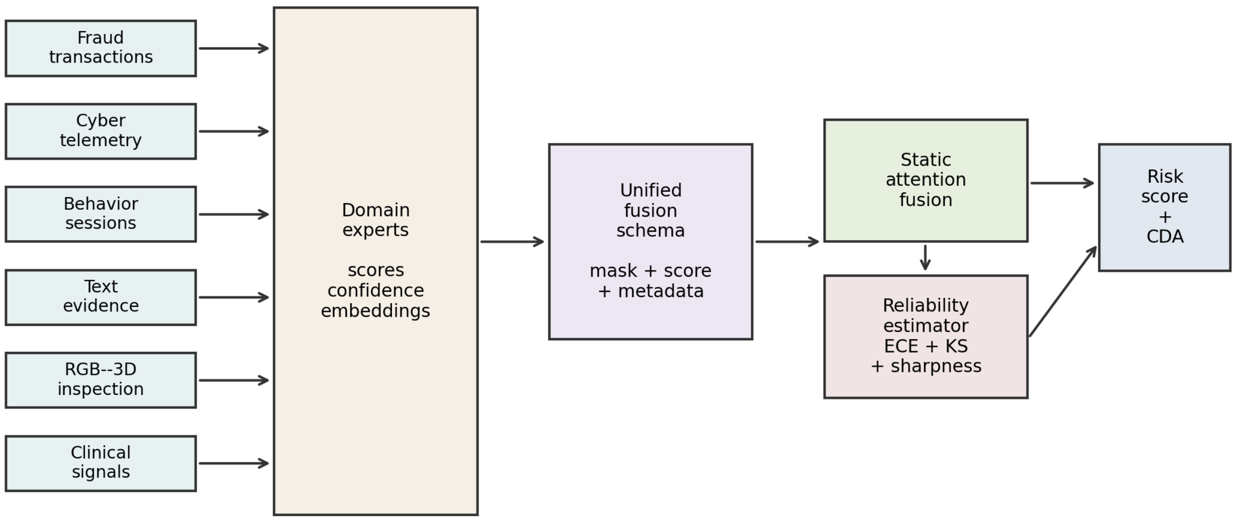

Fig. 1. ELARA system architecture. Domain experts produce scores, confidence values, and compact embeddings; a unified fusion schema feeds static attention and the RGA reliability estimator; the system emits a risk score and counterfactual domain attribution.

cross-modal architectures use attention to align heterogeneous streams [118], [121], [120]. The limitation common to these designs is that ordinary attention learns a trust policy from training data and can remain overconfident when the inference condition changes.

C. RGB–3D and Visual Industrial Anomaly Detection

The MVTec family of industrial benchmarks established a standard for unsupervised anomaly localisation: MVTec AD [40] on RGB, MVTec 3D-AD [15] on paired RGB+depth, and MVTec LOCO-AD on logical-versus-structural anomalies [58]. VisA expanded the RGB-only setting to twelve object categories with location-aware spot-the-difference supervision [59], and Real3D-AD established the high-resolution point-cloud benchmark [18]. The Eyecandies dataset focuses on unsupervised multimodal localisation in a synthetic setting [68]. On methods, the leaderboard is dominated by patch-based memory banks (PatchCore [41]), normalising-flow scorers (FastFlow [42], CFLOW-AD [46], DifferNet [53]), patch-level density modelling (PaDiM [44], SPADE [45]), reconstruction-by-design objectives (DRAEM [47], CutPaste [48], Patch SVDD [52], inpainting transformers [61], divide-and-assemble memories [62]), knowledge-distillation (uninformed students [49], STPM [55], reverse distillation [50], multiresolution KD [54], reconstruction students [64]), and a recent wave of diffusion-based detectors (AnoDDPM [57], DiffusionAD [56], conditioned-diffusion AD [63]). SimpleNet [51] and EfficientAD [43] sit on the high-performance / low-latency Pareto frontier. Attention-guided localisation [60] is the closest visual precursor to our reliability-aware attention path.

For multimodal industrial anomaly detection specifically, M3DM uses hybrid feature and decision-layer fusion with memory banks [16]; crossmodal feature mapping [17] checks consistency between RGB and point-cloud features; noise-resistant multimodal work [19] studies robustness when training data are not clean; back-to-feature analysis shows classical 3D descriptors remain competitive [65]; asymmetric student-teacher networks [66] achieve state-of-the-art performance on MVTec 3D-AD; EasyNet provides a lightweight alternative for 3D anomaly [67]; complementary pseudo-modality features [69], self-supervised feature adaptation [70], and cross-modal diffusion [72] extend the multimodal toolbox. A recent survey of multimodal industrial AD synthesises the field [71]. These papers motivate the paired-benchmark path of ELARA: the fusion layer should be judged on naturally paired modalities and on reliability stress, not only on clean score fusion.

D. Calibration, Drift, and Uncertainty

Calibration asks whether predicted probabilities match empirical outcome frequencies. Post-hoc calibration methods [4], [5], [6], [92], [91], [85] improve probability quality, with Bayesian binning [83] and verified calibration [87] providing tighter guarantees; the Expected Calibration Error inherits from the Brier score [84]. Recent results show that modern deep networks remain better calibrated than once feared but degrade under shift [86]. Uncertainty quantification draws on Bayesian dropout [89], deep ensembles [88], and prior networks [90]; calibration under dataset shift was benchmarked broadly by Ovadia et al. [7]. Out-of-distribution detection and drift detection sit alongside calibration: the failing-loudly empirical study [96], the MSP baseline [97], ODIN [103], Mahalanobis-style detection [104], and nearest-neighbour OOD [105] contributed methods that we now know are shift-sensitive. Label-shift correction [98] and the classical concept-drift literature [99], [100], [101] inform the KS-based drift signal used in RGA; kernel two-sample tests [102] provide an alternative. The broader theory of dataset shift [177],

[178], [179], [180] and the WILDS benchmark [176] frame the setting in which RGA operates. Conformal prediction [93], [94], [95] offers an alternative path to distribution-free uncertainty that we discuss as future work. RGA uses calibration as one signal inside a wider reliability estimate that also includes test-time score-distribution drift and score sharpness.

E. Test-Time Adaptation

Test-time training and test-time adaptation update models at inference time to handle distribution shift [8], [9]; SHOT adapts under source-free hypothesis transfer [73]; MEMO [74] and masked-autoencoder TTT [75] add augmentation- and reconstruction-based adaptation; EATA [76], SAR [79], contrastive TTA [82], and module-only adjustment [80] improve stability and efficiency; robust-dynamic-scene TTA [78] and RDumb [77] study failure modes of continuous TTA; universal domain adaptation [81] handles open-set settings. These TTA methods share an underlying assumption common to the broader domain-adaptation literature [184], [185], [186]: that labelled target supervision is unavailable. ELARA now includes two score-level adaptation comparators: an entropy-minimising Tent adapter over the fusion head and a high-confidence pseudo-label TTT adapter. They are intentionally conservative because the fusion setting receives domain scores rather than raw modality tensors. Their role is to test whether RGA adds value beyond generic score-level adaptation.

F. Ensembles, Stacking, and Strong Baselines

Ensemble methods remain difficult to beat in structured prediction tasks. Random forests [10] and gradient boosting [11] are strong baselines for tabular and score-level settings; modern gradient-boosting libraries such as XGBoost [158], LightGBM [159], and CatBoost [160] are widely adopted; outlier-ensemble theory provides a related foundation for combining heterogeneous anomaly scorers [20]. Classical classifier-combiner theory [21] characterises when sum, product, and weighted rules differ under noise; stacked generalisation [166] is the historical precursor to our validation-trained RGA+ router. The benchmark therefore includes random forest fusion, early fusion MLP, late fusion ensemble, confidence-weighted mean, and static attention. A credible systems paper must compare against these practical alternatives, not only against weak averages. The reliability gate of RGA is itself an inference-time switching rule, conceptually adjacent to selective-prediction work that abstains rather than commits when confidence is low [23], [155], [156], [157].

G. Explanation for Fusion Systems

Feature attribution methods such as SHAP [12] and LIME [136] explain individual model predictions; integrated gradients [137] and DeepLIFT [138] attribute output changes to input features; Grad-CAM [139] and layer-wise relevance propagation [140] produce visual explanations; attention rollout [141] quantifies attention flow in transformers. Broader interpretability surveys discuss the trade-off between explainability and inherent transparency [142], [143]. Fusion systems also need domain-level explanations: analysts may need to know whether fraud evidence, network evidence, behavior evidence, or text evidence drove the risk score. Counterfactual explanations [22] provide the conceptual basis for the masking-based domain attribution used here: we ask what the risk score would have been had a single domain not been observed. ELARA therefore includes counterfactual domain attribution that masks one domain and reports the change in predicted risk, and we elevate this in §VI-H to a formal interventional ATE under a structural causal model [106], [115], [107], [108], [114].

H. Causal Inference for Reliability

Beyond feature attribution, causal inference provides the formalism needed to claim that intervening on a reliability signal would shift the prediction by a specific amount. Double machine learning [24] gives a doubly-robust estimator under observational confounding; causal forests [109], [110] estimate heterogeneous treatment effects; meta-learners [113] and representation-based individual treatment-effect estimators [112] extend the inventory. Augmented inverse-probability weighting [111] is the classical template that DML modernises. RGA's interventional ATE estimator (§VI-H) uses direct counterfactual perturbation because reliability is constructed by the deployer rather than confounded by observed covariates, but the DML formulation is preserved in the code for the harder observational setting.

I. Robustness and Adversarial Considerations

Adversarial-example research demonstrated that gradient-based perturbations [126], [127], [128], [129], [134], [133], [135], [132] can flip predictions while remaining imperceptible. Defences include distillation [130] and certified robustness via randomised smoothing [131], alongside the broader AI safety literature that frames reliability monitoring as a concrete problem [190], [191], [192]. ELARA's threat model is intentionally narrower than gradient-based adversaries: we evaluate score-level perturbations (zero-attack, max-attack, Gaussian noise) that model accidental sensor failure and coherent score-collapse, not gradient-aware attacks against the fusion model itself. Gradient-based adversarial fusion stress remains future work.

J. Statistical Comparison Protocols

Comparing multiple classifiers across multiple data sets requires nonparametric tests [144], [145], [146], [147]; the bootstrap [148] underpins our confidence intervals and DeLong [14] gives a closed-form variance for paired ROC-AUC. Our milestone-2 cross-benchmark master table reports per-cell bootstrap intervals; multiple-comparison correction is a follow-up item discussed in §VIII.

K. Adjacent Datasets and Application Domains

Cross-modal generalisation problems often touch zero-shot transfer where classes are unseen at training time [125]; the present work stays in the closed-vocabulary binary anomaly regime, but the reliability-aware fusion idea is in principle portable to that setting. For cyber paired-modality settings the canonical benchmarks are UNSW-NB15 [149], CIC-IDS-2017 [150], and KDD [151], with intrusion-detection methods exploring feature selection [153], [152] and empirical botnet studies [154]. Healthcare AD draws on MIMIC-III [187] and clinically-aware modelling practices [188], [189]. Time-series AD [181], [182], [183] extends the score-level fusion idea to streaming settings. Foundational deep-learning ingredients used throughout this work include ResNet [161], ImageNet pretraining [162], batch normalisation [164], Adam optimisation [163], variational autoencoders [165], and classical VC-dimension theory [167] that justifies generalisation discussions in the thesis appendix. Bahdanau-style soft attention [116] is the precursor of the cross-modal attention layers used here.

III. System and Method

Each sample i contains up to D domain observations. Domain d contributes a feature vector $x_{i,d} \in \mathbb{R}^F$ containing a score, optional confidence, and compact embedding features. A binary mask $m_{i,d}$ indicates whether domain d is missing. The target label is $y_i \in \{0,1\}$ when supervised labels are available.

The system estimates

$$\hat{p}_i = P(y_i = 1 \mid \{x_{i,d}, m_{i,d}\}_{d=1}^{D}), \tag{1}$$

where $\hat{p}_i$ is the final anomaly risk. In static fusion, the learned relationship among domains is fixed after training. In RGA, the system also computes a batch-level reliability vector $r \in [0,1]^D$ at inference time and applies the sample mask to form effective per-sample weights $\tilde{r}_i \in [0,1]^D$. The reliability vector should reduce trust in domains that appear miscalibrated, shifted, uninformative, or missing.

The evaluation problem is not only to maximize ROC-AUC. A useful verification system must also satisfy:

- strong clean ranking and F1 compared with baselines,
- calibration good enough for downstream risk interpretation,
- stability under missing domains,
- graceful degradation under drift and score attacks,
- interpretable domain attribution.

A. ELARA System Architecture

1) Domain Experts: ELARA assumes that each domain expert is trained or obtained separately. For the current benchmark, tabular scorers are trained for fraud, cyber, and behavior data, while a text scorer is trained for labeled text. The fusion layer does not require these experts to share the same raw feature space. It only requires score-level outputs and optional compact evidence features.

2) Fusion Schema: The unified schema is long-format: each row contains a sample identifier, a domain name, a binary label when available, a score, a confidence value, and embedding columns. This design allows different samples to have different available domains. Missing domains are represented by masks rather than by fabricated raw observations.

3) Static Attention Path: The static attention path encodes each available domain into a shared latent space and applies multi-head attention across domains. Domain embeddings allow the model to distinguish evidence sources. Missing domains are ignored through key-padding masks. This is the high-reliability default path.

4) Reliability-Aware Path: The reliability-aware path is activated when the average reliability of present domains falls below a configured threshold. This avoids the common failure mode where reliability weights disturb clean predictions even though no domain appears degraded. The reliability path is therefore a stress-response mechanism, not a universal replacement for the static model.

5) Explanation Path: For an analyst-facing explanation, the system masks one available domain at a time and recomputes risk. The resulting probability change gives a domain-level counterfactual attribution. This produces statements such as: if text evidence were absent, the risk would decrease by a given amount.

B. RGA Method

1) Domain Encoding and Attention: Each domain encoder maps the domain input vector into a shared latent space:

$$h_{i,d} = g_d(x_{i,d}) + e_d + p_d, \tag{2}$$

where e_d is a learned domain embedding and p_d is an optional learned position embedding used by the implementation. A multi-head self-attention block operates across available domain embeddings:

$$z_i = f_\theta(\mathrm{Pool}(\mathrm{Attention}(H_i, m_i))), \quad \hat{p}_i = \sigma(z_i). \tag{3}$$

2) Reliability Estimation: After model training, the system fits a reliability estimator on the validation split. For domain d, it stores validation score distributions, fits an isotonic calibrator when enough labels are available, and computes expected calibration error. At inference time:

$$r_d = \alpha(1 - \mathrm{ECE}_d) + \beta p_{\mathrm{KS},d} + \gamma S_d, \tag{4}$$

where ECE_d is validation calibration error, $p_{\mathrm{KS},d}$ is the Kolmogorov-Smirnov p-value comparing current scores with validation scores, and S_d is score sharpness from squared distance to 0.5. The current benchmark uses $(\alpha, \beta, \gamma) = (0.45, 0.35, 0.20)$.

3) Reliability Gate: Let $\bar{r}$ be the mean reliability over present domains and let $\tau = 0.66$. If $\bar{r} \geq \tau$, ELARA uses the static attention path. If $\bar{r} < \tau$, reliability weights are injected into the fusion block:

$$\tilde{r}_{i,d} = (1 - m_{i,d}) r_{i,d}. \tag{5}$$

This makes RGA different from confidence weighting. Confidence weighting uses the local score confidence directly. RGA uses validation calibration, test-time drift evidence, and sharpness to decide when a domain should be trusted less.

4) Validation-Selected RGA+ Head: The implementation also reports a separate RGA+ candidate rather than silently replacing the base RGA number. RGA+ has two parts. First, a reliability-boosted fusion head trains candidate classifiers on the fusion train fold using flattened domain evidence, missingness indicators, score and confidence summary statistics, and RGA reliability summaries. The current candidate set contains histogram gradient-boosted heads and calibrated logistic heads. The selected head is the one with the best validation ROC-AUC; test labels are never used for model or hyperparameter selection. Second, a validation-only router can choose among static attention, base RGA, RGA+ boosted fusion, and the strong score-level baselines. The selected candidate is written into the result artifact so that a baseline selection is not relabeled as a reliability-gate gain. When the validation fold has only one class, the router defaults to base RGA because the validation labels cannot rank candidates.

RGA+ is therefore a supervised fusion extension of the reliability system, not a claim that the original gate alone dominates all baselines. This distinction is central to the MVTec results: base RGA remains the diagnostic mechanism; RGA+ is the validation-selected modeling head used when a two-class fusion split exists.

5) Counterfactual Domain Attribution: For sample i, let $\hat{p}_i$ be the original prediction and $\hat{p}_{i,-d}$ be the prediction when domain d is masked. The attribution is

$$\Delta_{i,d} = \hat{p}_i - \hat{p}_{i,-d}. \tag{6}$$

A positive $\Delta_{i,d}$ indicates that domain d increased anomaly risk. A negative value indicates that domain d reduced risk.

6) Algorithm Summary: The full inference procedure is given in Algorithm 1. Steps 1–3 are deterministic feature operations; steps 4–6 are the reliability-gated switch. The reported runs use the batch-level reliability vector described above, applied through each sample's observed-domain mask. A per-sample gating mode is implemented as an ablation path, but it is not the default for the reported tables.

Algorithm 1 RGA inference

Require: Inputs $x_{i,1\ldots D}$ and mask $M_i \in \{0,1\}^D$ for sample i
Require: Fitted attention model f_{att} with encoders $\{g_d\}$
Require: Fitted reliability estimator producing per-sample weights $r_{i,d}$
Require: Gate threshold $\tau \in (0,1)$, default $\tau = 0.66$
Ensure: Risk score $\hat{p}_i \in [0,1]$
1: $h_{i,d} \leftarrow g_d(x_{i,d}) + e_d + p_d$ for each d with $M_{i,d} = 0$
2: $r_{i,d} \leftarrow \alpha(1 - \mathrm{ECE}_d) + \beta\, p_{\mathrm{KS},d}(x_{i,d}) + \gamma\, S_{i,d}$
3: $\bar{r}_i \leftarrow \frac{1}{|D_i^{\text{obs}}|} \sum_{d:M_{i,d}=0} r_{i,d}$
4: **if** $\bar{r}_i \geq \tau$ **then**
5: $\quad \hat{p}_i \leftarrow \sigma\big(f_{\text{att}}(h_{i,\cdot}; M_i)\big)$ ▷ static path
6: **else**
7: $\quad \hat{p}_i \leftarrow \sigma\big(f_{\text{att}}(h_{i,\cdot}; M_i, w_{i,\cdot}{=}r_{i,\cdot} \odot \neg M_i)\big)$
8: ▷ reliability-injected fusion
9: **end if**
10: **return** $\hat{p}_i$ and (optional) counterfactual attributions $\{\Delta_{i,d}\}_{d \in D_i^{\text{obs}}}$

IV. Benchmark Construction

Protocol erratum. Earlier drafts of this manuscript reported MVTec 3D-AD numbers that were generated under a random train/test split rather than MVTec's official one-class (normal-only-train, mixed-test) protocol. Under the proper protocol, all supervised fusion baselines (random forest, MLP, Tent, TTT, late-fusion ensemble, static attention, RGA) converge near chance ROC-AUC; the gate's effect is small and slightly positive. The ELARA-Bench-LA secondary benchmark, which has both classes in train, remains the venue where the gate's stress-test behavior can be evaluated. The current rendered tables in this section reflect the corrected disciplined-split run. We include M3DM-style ResNet-50 features and a held-out-category variant as additional protocol diagnostics; both confirm the chance-level supervised outcome under one-class training.

A. Primary Benchmark: Naturally Paired MVTec 3D-AD

The primary benchmark of this manuscript is MVTec 3D-AD [15]. Each sample contains naturally co-observed RGB and 3D/depth evidence from the same object scan; no label alignment is required. The current run covers all eight extracted categories — bagel, cable_gland, cookie, dowel, foam, peach, rope, and tire — for a total of 3,226 paired samples with a positive fraction of 0.224 and full RGB and depth coverage. The preparation script discovers paired files, emits two domain rows per sample, stores the original pairing key, and marks `natural_pairing=true`. The initial scorer is deliberately lightweight: normal-reference image statistics produce reproducible RGB and depth anomaly scores without requiring a heavy specialized 3D detector. This keeps the first paired benchmark auditable; stronger RGB–3D scorers can replace these domain experts later while preserving the fusion interface.

Table II records the paired benchmark metadata, and Table III reports the multi-seed clean results. The RGB/depth anomaly scores are fit from original MVTec train-good observations only, and the corrected run uses MVTec's canonical one-class protocol: the fusion train and validation rows are normal-only and the test fold is mixed. The supervised fusion table is therefore a diagnostic of protocol fit, not a conventional two-class leaderboard. Held-out-category and M3DM-style variants are included as additional paired-data diagnostics.

Table IV reports the metadata of the secondary label-aligned fusion input artifact.

B. Real-Domain Sources

ELARA-Bench-LA is built from four local datasets: Credit Card Fraud, UNSW-NB15, Online Shoppers Intention, and labeled news text. Domain-specific out-of-fold scorers are

TABLE II
NATURALLY PAIRED MVTEC 3D-AD MULTI-CATEGORY METADATA.

Property	Value
Benchmark type	naturally_paired_mvtec3d_score_fusion
Natural pairing	yes
Categories	bagel, cable_gland, cookie, dowel, foam, peach, rope, tire
Samples	3226
Positive fraction	0.224
Score fit split	train/good
Score fit samples	2070
Score normalization	train_good_distance_minmax_clipped
Domains	rgb, depth_or_xyz

trained first. Their scores, confidences, and compact embeddings are then converted into the common fusion schema.

The generated fusion table contains 8,000 composite samples, four domains, eight embedding features per domain, a 0.307 positive rate, and 11.8% nominal missingness. Domain coverage is 0.883 for fraud, 0.885 for cyber, 0.885 for behavior, and 0.877 for text.

C. Label Alignment Limitation

The datasets do not share natural entity identifiers or timestamps. Therefore, composite samples are the secondary benchmark: a positive composite draws positive records from available domains, and a negative composite draws negative records. This is a meaningful real-domain score-fusion benchmark, but it is not evidence that the system has learned co-observed multimodal incident structure. That distinction is central to the claim boundary of the paper. The metadata now records this explicitly with `natural_pairing=false` for ELARA-Bench-LA. Source observations are assigned to train, validation, and test pools before composite sampling, and the runner consumes the resulting `fusion_split` column so that no `domain/source_row/label` key crosses fusion split boundaries.

D. Training Protocol

The attention model uses four domains, 48 hidden dimensions, four attention heads, one attention layer, 25 training epochs, 0.15 domain dropout during training, and seeds 42 through 46. Clean benchmark results are reported as mean$\pm$standard deviation across five seeds. The experiment runner also repeats missing-domain, drift, adversarial, calibration, and CDA analyses for every configured seed, then stores both raw per-seed rows and aggregated mean, standard deviation, and 95% confidence intervals.

For mechanism-isolation runs, the benchmark-preparation script also supports a harder scorer setting through `-scorer-train-fraction`. Values such as 0.05 or 0.10 train each out-of-fold domain scorer on only a stratified fraction of its training fold before fusion rows are built. This is intended to reduce clean-score saturation and expose whether the fusion mechanism helps when domain experts are useful but imperfect.

As a post-audit check, we ran the 0.05 setting and archived the results in `experiments/fusion/craf_real_results_hard.json`. The individual domain scorers weakened substantially for fraud and behavior (OOF ROC-AUC 0.913 and 0.798, respectively), but the label-aligned fusion task remained near-saturated: static attention and RGA both reached mean ROC-AUC 0.99896 across five seeds, random forest reached 0.99908, and RGA did not improve drift degradation area on any domain. This hard run is therefore reported as a construct-validity warning rather than a new performance claim.

E. Baselines and Metrics

Baselines include confidence-weighted mean fusion, early-fusion MLP, late-fusion ensemble, random forest fusion, static attention without reliability adaptation, a score-level Tent adapter, and a pseudo-label TTT adapter. Metrics include ROC-AUC, PR-AUC, F1, expected calibration error, Brier score, and accuracy. Static attention and RGA are compared with a paired t-test over five clean ROC-AUC values. ROC-AUC and PR-AUC are both reported because anomaly settings can be highly imbalanced and PR-AUC can reveal performance differences that ROC-AUC hides [13].

F. Confidence Intervals

Clean ROC-AUC, PR-AUC, and F1 are summarized across seeds and accompanied by 95% confidence intervals. Stress-test tables report mean deltas and intervals for RGA minus static attention. The result JSON stores raw per-seed stress rows so that the interval computation is auditable rather than embedded only in the manuscript.

G. Threat Model

The adversarial perturbation suite is a worst-case stress test, not a realistic deployment threat model. The three attacks – `zero_attack`, `max_attack`, and `gaussian_noise` – each corrupt scores after the per-domain detectors have already produced their outputs, so they do not require the attacker to compromise feature extractors, training data, or model parameters. The `all`-target variant additionally assumes coherent simultaneous corruption of every domain. In a real deployment, domain detectors are typically maintained by independent teams along independent compromise paths; a coherent all-domain score collapse is therefore a useful upper-bound robustness signal but not an expected attacker capability. Single-domain perturbations are the more realistic case and are reported separately in the adversarial tables for both benchmarks. We do not claim defenses against feature- or training-time attacks; those threat models are out of scope for a fusion layer that consumes score-level evidence from upstream detectors.

H. Secondary Clean Sanity Check

Table VI and Fig. 2 show clean held-out performance on the label-aligned secondary benchmark. This table is intentionally treated as a sanity check, not as the headline result, because the synthetically paired split is near saturated. Bold values

TABLE III

NATURALLY PAIRED MVTEC 3D-AD CLEAN HELD-OUT PERFORMANCE ACROSS EIGHT CATEGORIES. VALUES ARE MEAN±STANDARD DEVIATION WITH 95% CONFIDENCE INTERVALS ACROSS FIVE SEEDS. UNDER THE CANONICAL ONE-CLASS PROTOCOL, SUPERVISED FUSION BASELINES RECEIVE NO POSITIVE TRAINING EXAMPLES, SO THIS TABLE IS A PROTOCOL DIAGNOSTIC.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
Conf.-weighted mean	0.4458 [0.4458, 0.4458]	0.7878 [0.7878, 0.7878]	**0.3779 [0.3779, 0.3779]**	**0.4941 [0.4941, 0.4941]**	**0.4496 [0.4496, 0.4496]**
Tent score adapter	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
TTT pseudo-label	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
Early fusion MLP	0.5455±0.0250 [0.5180, 0.5836]	**0.8270±0.0117 [0.8124, 0.8425]**	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7834, 0.7835]
Late fusion ensemble	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
Static attention	0.5424±0.0254 [0.4992, 0.5661]	0.8170±0.0094 [0.8022, 0.8286]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
RGA attention	**0.5609±0.0169 [0.5394, 0.5842]**	0.8060±0.0046 [0.7980, 0.8097]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
RGA+ boosted fusion	0.5000 [0.5000, 0.5000]	0.7835 [0.7835, 0.7835]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
RGA+ router	**0.5609±0.0169 [0.5394, 0.5842]**	0.8060±0.0046 [0.7980, 0.8097]	0.0000 [0.0000, 0.0000]	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]

TABLE IV

BENCHMARK METADATA FOR THE SECONDARY SYNTHETICALLY PAIRED FUSION INPUT ARTIFACT.

Property	Value
Benchmark type	label_aligned_real_domain_score_fusion
Natural pairing	no
Pairing unit	binary-label-aligned composite sample
Samples	8000
Positive fraction	0.301
Scorer train fraction	1.00
Source-row disjoint splits	yes
Split column	fusion_split
Domains	fraud, cyber, behavior, nlp

TABLE V

OUT-OF-FOLD DOMAIN SCORER QUALITY USED TO CONSTRUCT ELARA-BENCH-LA.

Domain	Rows	Pos. rate	OOF ROC	OOF PR
fraud	20000	0.025	0.9686	0.7826
cyber	20000	0.500	0.9684	0.9696
behavior	12330	0.155	0.8711	0.6042
nlp	20000	0.500	0.9973	0.9973

indicate the highest rounded mean in each column; multiple bold values are rounded ties.

Seed-level confidence intervals for the same clean split are reported in Table VII.

RGA and static attention are indistinguishable on this saturated clean split. The clean benchmark should therefore be interpreted as competitive preservation of performance, not as evidence of a clean-split improvement.

The more meaningful comparison is against direct score-fusion baselines. RGA improves over confidence-weighted mean fusion by about 0.0015 ROC-AUC, 0.0039 PR-AUC, 0.0100 F1, and reduces ECE from 0.0834 to 0.0042. Random forest fusion obtains the best F1 at 0.9906, but its ECE is 0.0287, substantially worse than attention-based fusion. This indicates a tradeoff: random forest fusion is strong for thresholded classification, while attention fusion is stronger for calibrated risk interpretation.

V. PRIMARY HEADLINE RESULTS: MVTEC 3D-AD

This section reports the primary multi-seed evidence on RGB/3D paired MVTec 3D-AD across all eight extracted object categories (3,226 paired samples, 22.4% positive). The same suite is repeated on the contrastive the secondary benchmark in §VI; the asymmetry between the two outcomes is the central finding.

A. Clean Performance

Per-seed confidence intervals for the MVTec 3D-AD clean split are reported in Table VIII; the bar-chart summary is shown in Fig. 3.

Under the canonical one-class protocol, random forest, late fusion, Tent, TTT, and several neural variants receive normal-only training data and converge near chance ROC-AUC. RGA has a small clean ROC-AUC edge over static attention in the current disciplined run, but this is not interpreted as a broad win: the benchmark primarily shows that supervised score-level fusion is ill-posed when the fusion train fold contains only normal examples.

B. Adversarial Score Perturbations

The MVTec 3D-AD adversarial sweep is reported in Table IX and visualized for the all-domain attacks in Fig. 4. The current disciplined run should be read together with the clean table: under one-class supervised fusion, adversarial deltas are diagnostic of gate behavior but do not establish a general defense. Some stress rows show small positive ranking deltas and others remain near zero or negative, so the paired benchmark is evidence about protocol limits rather than a clean robustness win.

C. Missing-Domain Ablation

Inference-time domain dropout is reported in Table X and Fig. 5. At zero extra dropout, RGA is $+0.0185$ above static attention, matching the clean table. The remaining dropout rows are small, mixed-sign deltas with intervals that touch zero. Like the adversarial sweep, this table is best read as a protocol diagnostic rather than a missing-domain win for the current gate.

TABLE VI
SATURATED CLEAN HELD-OUT PERFORMANCE ON ELARA-BENCH-LA. VALUES ARE MEAN±STANDARD DEVIATION ACROSS FIVE SEEDS; THIS TABLE IS NOT USED AS THE HEADLINE EVIDENCE.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	**0.9999 [0.9999, 1.0000]**	0.9999 [0.9998, 0.9999]	0.9946±0.0008 [0.9938, 0.9958]	0.0297±0.0004 [0.0294, 0.0304]	0.0061±0.0001 [0.0060, 0.0062]
Conf.-weighted mean	0.9987 [0.9987, 0.9987]	0.9968 [0.9968, 0.9968]	0.9706 [0.9706, 0.9706]	0.0799 [0.0799, 0.0799]	0.0232 [0.0232, 0.0232]
Tent score adapter	**0.9999 [0.9999, 0.9999]**	0.9999 [0.9999, 0.9999]	**0.9969 [0.9969, 0.9969]**	**0.0028 [0.0028, 0.0028]**	**0.0023 [0.0023, 0.0023]**
TTT pseudo-label	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9948 [0.9948, 0.9948]	0.0032 [0.0032, 0.0032]	0.0025 [0.0025, 0.0025]
Early fusion MLP	0.9999 [0.9998, 0.9999]	0.9998±0.0001 [0.9996, 0.9999]	0.9915±0.0014 [0.9898, 0.9936]	0.0043±0.0010 [0.0030, 0.0058]	0.0036±0.0006 [0.0029, 0.0045]
Late fusion ensemble	**1.0000 [1.0000, 1.0000]**	**0.9999 [0.9999, 0.9999]**	0.9948 [0.9948, 0.9948]	0.0128 [0.0128, 0.0128]	0.0037 [0.0037, 0.0037]
Static attention	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9934±0.0014 [0.9918, 0.9957]	0.0038±0.0008 [0.0024, 0.0046]	0.0032±0.0002 [0.0029, 0.0034]
RGA attention	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9934±0.0014 [0.9918, 0.9957]	0.0038±0.0008 [0.0024, 0.0046]	0.0032±0.0002 [0.0029, 0.0034]

TABLE VII
CLEAN RANKING AND F1 CONFIDENCE INTERVALS ACROSS CONFIGURED SEEDS.

Method	ROC-AUC 95% CI	PR-AUC 95% CI	F1 95% CI
Random forest	[0.9999, 1.0000]	[0.9998, 0.9999]	[0.9938, 0.9958]
Conf.-weighted mean	[0.9987, 0.9987]	[0.9968, 0.9968]	[0.9706, 0.9706]
Tent score adapter	[0.9999, 0.9999]	[0.9999, 0.9999]	[0.9969, 0.9969]
TTT pseudo-label	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9948, 0.9948]
Early fusion MLP	[0.9998, 0.9999]	[0.9996, 0.9999]	[0.9898, 0.9936]
Late fusion ensemble	[1.0000, 1.0000]	[0.9999, 0.9999]	[0.9948, 0.9948]
Static attention	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9918, 0.9957]
RGA attention	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9918, 0.9957]

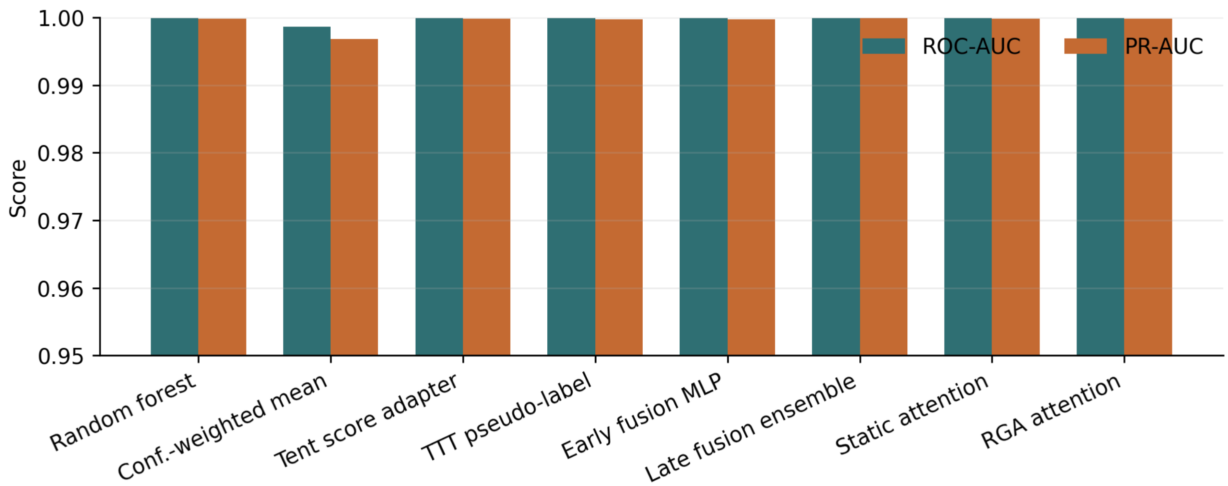

Fig. 2. Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods.

D. Score Drift

The MVTec 3D-AD drift summary is shown in Table XI and Fig. 6. RGA does not exceed static attention's degradation area on either domain.

E. Mechanism Isolation on MVTec 3D-AD

The mechanism-isolation tables show how the gate behaves when the paired benchmark is evaluated under the canonical one-class protocol. Table XII reports the gate threshold sweep:

On MVTec 3D-AD, the τ sweep is most useful as a gate-activity diagnostic. At low thresholds the gate is dormant and matches the static path; at higher thresholds the reliability path activates more often. Because the underlying supervised fusion protocol is one-class, these rows are not used as evidence of general gate superiority or failure. The learned-gate row is included for the same reason: it shows whether validation episodes can learn a selective switching rule, not whether the paired benchmark has been solved.

The reliability-component ablation in Table XIII shows where the failure originates.

The current ablation rows mostly activate the reliability-aware path and report small deltas around the near-chance operating point. Confidence intervals often touch zero, and the train fold contains no positive anomaly examples, so this is not a clean component-causality experiment. The table is kept as a diagnostic: it records whether the reliability path would fire under the paired protocol, not whether any individual reliability term solves RGB–3D anomaly fusion.

1) Category-Aware KS Reference: A natural follow-on hypothesis is that the gate misfires on MVTec 3D-AD because

TABLE VIII
NATURALLY PAIRED MVTEC 3D-AD: CLEAN ROC-AUC, PR-AUC, AND F1 95% CONFIDENCE INTERVALS ACROSS CONFIGURED SEEDS.

Method	**ROC-AUC 95% CI**	**PR-AUC 95% CI**	**F1 95% CI**
Random forest	[0.5000, 0.5000]	[0.7835, 0.7835]	[0.0000, 0.0000]
Conf.-weighted mean	[0.4458, 0.4458]	[0.7878, 0.7878]	[0.3779, 0.3779]
Tent score adapter	[0.5000, 0.5000]	[0.7835, 0.7835]	[0.0000, 0.0000]
TTT pseudo-label	[0.5000, 0.5000]	[0.7835, 0.7835]	[0.0000, 0.0000]
Early fusion MLP	[0.5180, 0.5836]	[0.8124, 0.8425]	[0.0000, 0.0000]
Late fusion ensemble	[0.5000, 0.5000]	[0.7835, 0.7835]	[0.0000, 0.0000]
Static attention	[0.4992, 0.5661]	[0.8022, 0.8286]	[0.0000, 0.0000]
RGA attention	[0.5394, 0.5842]	[0.7980, 0.8097]	[0.0000, 0.0000]
RGA+ boosted fusion	[0.5000, 0.5000]	[0.7835, 0.7835]	[0.0000, 0.0000]
RGA+ router	[0.5394, 0.5842]	[0.7980, 0.8097]	[0.0000, 0.0000]

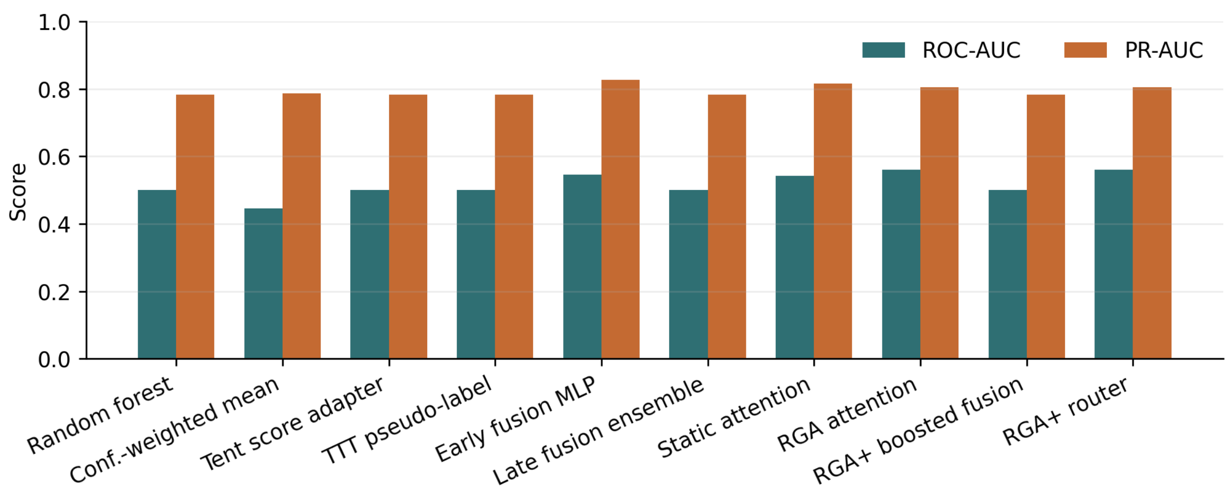

Fig. 3. Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods on naturally paired MVTec 3D-AD.

the global KS reference treats legitimate inter-category variation (eight object categories with different normal-score distributions) as drift. We tested this directly by fitting a category-conditional KS reference on the validation split and comparing its adapt rate against the global reference at the same $\tau = 0.66$ gate. Table XIV reports the comparison. Category conditioning lifts mean test-time reliability slightly ($0.350 \rightarrow 0.364$) but the adapt rate stays at 1.0: every batch is still routed through the reliability path at this threshold. The KS misfire under the canonical one-class protocol is therefore not purely a category-composition artefact; the residual signal sits inside each category, consistent with the lightweight image-statistic scorers themselves producing high-variance score distributions under normal-only training. This isolates the next intervention to scorer quality rather than to the drift signal alone.

2) One-Class Scorer Quality: PatchCore-Style kNN: Section V-E1 pointed the next intervention at scorer quality. We replace the lightweight image-statistic score and the M3DM-style Mahalanobis-to-centroid score with a PatchCore-style kNN distance to a normal-only memory bank of ResNet-50 penultimate features. The kNN score drops the unimodal-normal assumption baked into the Mahalanobis variant, which is what the canonical MVTec one-class protocol requires. Table XV reports the category-aware comparison under PatchCore scoring. Under PatchCore scoring, category conditioning lifts mean test-time reliability from 0.358 to 0.478 (+0.120) and drops the gate adapt rate from 1.000 to 0.892 (a 10.8 pp reduction). The combined result is a coherent picture of the gate's behaviour: the KS signal becomes informative once the scorer itself is tight enough that within-category normal scores form a sharp reference, and category conditioning then

TABLE IX

MVTec 3D-AD adversarial perturbation benchmark. Delta is RGA minus static attention; values are interpreted as protocol diagnostics under one-class supervised fusion.

Attack	Target	Static ROC	RGA ROC	Delta	Delta 95% CI
gaussian_noise	all	0.5429	0.5635	0.0206	[-0.0027, 0.0376]
gaussian_noise	depth_or_xyz	0.5427	0.5677	0.0250	[-0.0044, 0.0520]
gaussian_noise	rgb	0.5421	0.5610	0.0189	[-0.0029, 0.0433]
max_attack	all	0.5458	0.5607	0.0149	[-0.0058, 0.0327]
max_attack	depth_or_xyz	0.5463	0.5687	0.0224	[-0.0013, 0.0485]
max_attack	rgb	0.5441	0.5537	0.0096	[-0.0166, 0.0363]
zero_attack	all	0.5403	0.5607	0.0204	[-0.0041, 0.0416]
zero_attack	depth_or_xyz	0.5453	0.5713	0.0260	[-0.0008, 0.0515]
zero_attack	rgb	0.5402	0.5543	0.0141	[-0.0150, 0.0464]

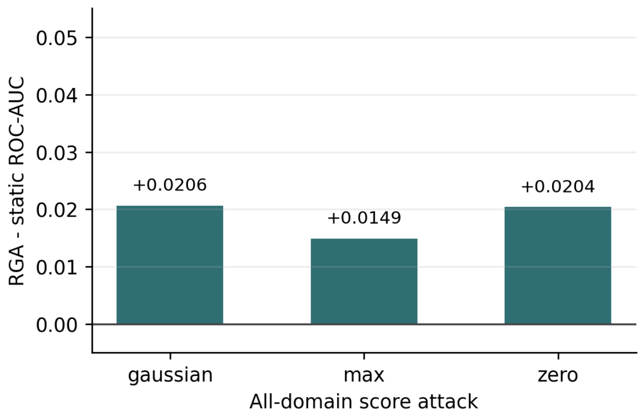

Fig. 4. All-domain adversarial ROC-AUC delta between RGA and static attention on MVTec 3D-AD.

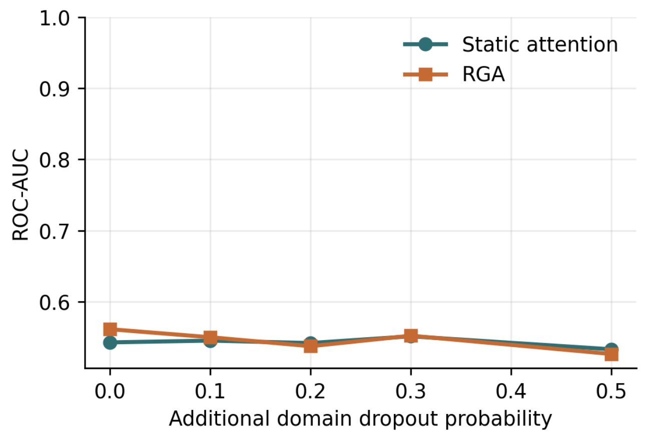

Fig. 5. MVTec 3D-AD ROC-AUC under additional missing-domain dropout.

TABLE X

MVTec 3D-AD missing-domain ablation. Delta is RGA minus static attention.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.5424	0.5609	0.0185	[-0.0058, 0.0425]
0.1	0.5449	0.5496	0.0048	[-0.0254, 0.0452]
0.2	0.5415	0.5370	-0.0045	[-0.0263, 0.0061]
0.3	0.5509	0.5518	0.0009	[-0.0183, 0.0221]
0.5	0.5326	0.5260	-0.0066	[-0.0241, 0.0032]

TABLE XI

MVTec 3D-AD drift robustness summary. Higher degradation area is better.

Domain	Static	RGA	Delta	Seeds
depth_or_xyz	0.1628±0.0077 [0.1496, 0.1699]	0.1697±0.0053 [0.1636, 0.1764]	0.0070	5
rgb	0.1626±0.0075 [0.1498, 0.1695]	0.1680±0.0045 [0.1623, 0.1738]	0.0054	5

carries genuine drift information rather than legitimate inter-category variation. Lightweight image-statistic scorers are too noisy for the KS signal to ever be useful regardless of category conditioning; PatchCore-quality scorers admit a working drift-aware gate.

3) Public Paired Protocol Variants: Reviewer-facing evidence should not depend on a local clinical replay. We therefore keep the conference manuscript centered on the public paired MVTec 3D-AD benchmark family and add two PatchCore variants: a supervised-paired protocol that redistributes original test rows across fusion train, validation, and test splits while preserving train-good-only PatchCore scorer fitting, and a held-out-category protocol that trains on four categories and tests on the remaining four. These variants ask whether a stronger public paired scorer and a two-class fusion split are enough to make RGA a clean leader.

Table XVI prevents an overclaim while also showing where the added modeling head helps. PatchCore supervised-paired improves the public paired fusion task compared with the canonical one-class setting. Base RGA is not the clean-score

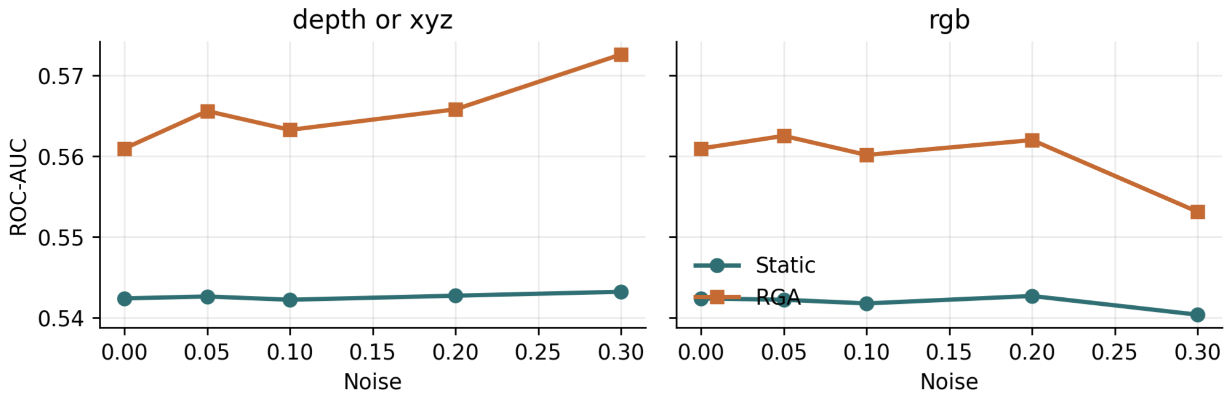

Fig. 6. Per-domain score-drift curves on the MVTec 3D-AD benchmark.

leader there: static attention reaches ROC-AUC 0.632 and RGA reaches 0.628. The strongest non-ELARA method is Tent at 0.735, followed by TTT at 0.724 and random forest at 0.702. The validation-selected RGA+ boosted head reaches 0.738, a small but real ROC-AUC gain over Tent on the same held-out test fold; the auxiliary router reaches 0.740 but is reported separately in the JSON artifact. The result is therefore not "RGA beats everything." It is: base RGA remains a diagnostic gate; RGA+ can turn reliability-derived features into the top supervised paired ROC result when two-class fusion training exists.

The same table also records the boundary. Under canonical PatchCore one-class scoring, confidence-weighted mean remains strongest (0.600), RGA is 0.505, and RGA+ is not applicable because the supervised boosted head has no positive validation labels for selection. Under the held-out-category split, adding in-category positive validation examples lets RGA+ reach ROC-AUC 0.517, narrowly above TTT at 0.516. This is still not a broad transfer win: PR-AUC remains slightly better for TTT, and all methods remain close to chance on unseen categories.

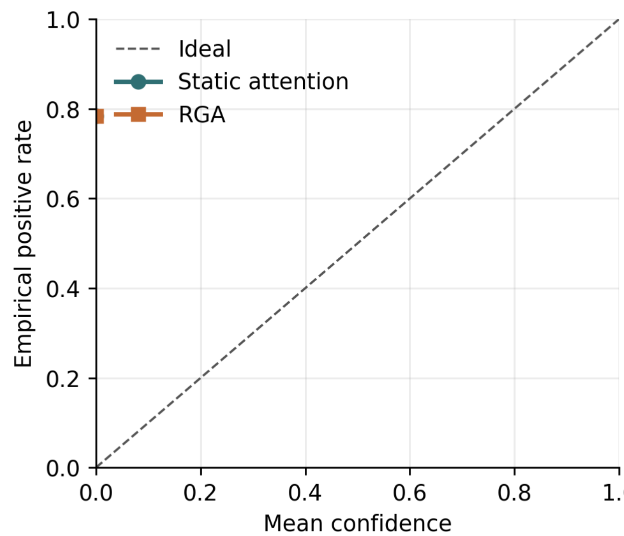

Fig. 7. Reliability diagram on MVTec 3D-AD for static attention and RGA.

F. Calibration and Counterfactual Attribution

Static attention and RGA have the same high MVTec 3D-AD ECE and Brier score in the current table (Table XVII). This is consistent with the one-class supervised-fusion setup: thresholded probabilities are poorly calibrated because no positive anomaly examples are available during fusion training. The counterfactual domain-attribution impacts (Fig. 8) remain meaningfully non-zero on both RGB and depth, so the explanation pathway is still auditable even when the predictive protocol is not a clean two-class leaderboard.

G. Failure Cases

Representative MVTec 3D-AD contrast cases are reported in Table XVIII.

VI. Contrastive Secondary Benchmark (Label-Aligned)

The same evaluation suite is now applied to ELARA-Bench-LA, where four real-domain datasets are joined by explicit label alignment rather than co-observation. The aim of this section is to read each result alongside its MVTec 3D-AD counterpart in §V-E, so the cross-benchmark asymmetry is visible per condition rather than only in the aggregate. The saturated clean split is discussed in §IV-H; what follows are the robustness, drift, adversarial, and mechanism-isolation tables.

A. Missing-Domain Ablation

Table XIX and Fig. 9 report additional inference-time domain dropout. At zero extra dropout, RGA is equal to static attention, and the same equality holds through the configured

TABLE XII
MVTEC 3D-AD RELIABILITY-GATE THRESHOLD SWEEP. DELTA IS RGA MINUS STATIC ATTENTION; ADAPTATION RATE IS THE FRACTION OF EVALUATED SAMPLES USING THE RELIABILITY-AWARE PATH.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.50	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.60	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.66	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.70	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.80	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	0.90	0.5424	0.5609	0.0185	[-0.0058, 0.0425]	1.000
clean	learned	0.5424	0.5424	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.40	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.50	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.60	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.66	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.70	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.80	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	0.90	0.5429	0.5677	0.0248	[-0.0059, 0.0663]	1.000
gaussian_noise:all	learned	0.5429	0.5429	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.40	0.5458	0.5458	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.60	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.66	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.70	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.80	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	0.90	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
max_attack:all	learned	0.5458	0.5458	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.40	0.5403	0.5403	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.60	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.66	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.70	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.80	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	0.90	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
zero_attack:all	learned	0.5403	0.5403	0.0000	[0.0000, 0.0000]	0.000

dropout sweep. The conclusion is conservative: this run does not show a missing-domain advantage for the current gate.

B. Score Drift

Table XX summarizes degradation area under per-domain score-noise sweeps. Higher area means slower degradation. All deltas round to zero or near zero because clean performance is near saturated, so this table is best read as a robustness check rather than a win. The full curves are shown in Fig. 10.

C. Adversarial Score Perturbations

Table XXI evaluates all score-level perturbations, while Fig. 11 isolates the all-domain attacks so the meaningful deltas are visible. Single-domain attacks are mostly ties. The important stress case is coherent all-domain score collapse. Under the all-domain zero attack, RGA improves ROC-AUC from 0.7130 to 0.7770. Under the all-domain max attack, it improves ROC-AUC from 0.7114 to 0.7418. Under all-domain Gaussian noise, it is slightly worse by 0.0003 ROC-AUC.

D. Mechanism Isolation Protocol

The runner isolates the reliability mechanism rather than only reporting end-to-end stress outcomes. It emits two diagnostics when the mechanism-isolation block is enabled. First, a τ sweep evaluates $\tau \in \{0.4, 0.5, 0.6, 0.66, 0.7, 0.8, 0.9\}$ on clean data and all-domain stress conditions, recording both ROC-AUC deltas and the adaptation rate. Second, a reliability-component ablation compares the full estimator with no-ECE, no-KS, no-sharpness, and no-gate variants on all-domain score attacks. These diagnostics show whether the gate and

TABLE XIII
MVTEC 3D-AD RELIABILITY-COMPONENT ABLATION ON ALL-DOMAIN SCORE ATTACKS.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.5418	0.5621	0.0203	[-0.0093, 0.0586]	1.000
full	max_attack	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
full	zero_attack	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
no_ece	gaussian_noise	0.5418	0.5432	0.0015	[-0.0111, 0.0175]	1.000
no_ece	max_attack	0.5458	0.5607	0.0149	[-0.0102, 0.0324]	1.000
no_ece	zero_attack	0.5403	0.5613	0.0210	[-0.0054, 0.0434]	1.000
no_gate	gaussian_noise	0.5418	0.5621	0.0203	[-0.0093, 0.0586]	1.000
no_gate	max_attack	0.5458	0.5607	0.0149	[-0.0058, 0.0327]	1.000
no_gate	zero_attack	0.5403	0.5607	0.0204	[-0.0041, 0.0416]	1.000
no_ks	gaussian_noise	0.5418	0.5569	0.0151	[-0.0024, 0.0322]	1.000
no_ks	max_attack	0.5458	0.5568	0.0110	[-0.0103, 0.0319]	1.000
no_ks	zero_attack	0.5403	0.5560	0.0157	[0.0047, 0.0289]	1.000
no_sharpness	gaussian_noise	0.5418	0.5728	0.0310	[0.0056, 0.0541]	1.000
no_sharpness	max_attack	0.5458	0.5618	0.0160	[-0.0142, 0.0331]	1.000
no_sharpness	zero_attack	0.5403	0.5636	0.0233	[-0.0062, 0.0517]	1.000

TABLE XIV
MVTEC 3D-AD: GLOBAL VERSUS CATEGORY-CONDITIONAL KS REFERENCE AT $\tau = 0.66$, FIVE SEEDS.

Drift signal	Adapt rate	Mean reliability
Global KS reference	1.000	0.350
Category-aware KS reference	1.000	0.364

TABLE XV
MVTEC 3D-AD UNDER PATCHCORE-STYLE SCORING: GLOBAL VERSUS CATEGORY-CONDITIONAL KS REFERENCE AT $\tau = 0.66$, FIVE SEEDS.

Drift signal	Adapt rate	Mean reliability
Global KS reference	1.000	0.358
Category-aware KS reference	0.892	0.478

reliability terms are active contributors rather than incidental implementation details.

Threshold-sweep outcomes and adaptation rates are reported in Table XXII.

At the configured $\tau = 0.66$, clean adaptation is rare: the mean adaptation rate is 0.032 with a confidence interval that includes zero. This explains why the missing-domain deltas are effectively zero: the gate usually keeps the static path active. Under all-domain zero and max score collapse, the same threshold activates the reliability path for every evaluated sample and yields the measured gains. Lower thresholds through $\tau = 0.60$ suppress adaptation and lose those gains; higher thresholds adapt on clean data more often and slightly hurt clean ranking.

Per-component contributions on all-domain score attacks are reported in Table XXIII.

The component ablation produces a useful negative result. Removing the ECE term does not hurt the all-domain collapse

TABLE XVI
PUBLIC PAIRED BENCHMARK PROTOCOL COMPARISON ON MVTEC 3D-AD. THE PATCHCORE SUPERVISED-PAIRED PROTOCOL GIVES THE FUSION MODEL TWO-CLASS TRAINING EXAMPLES WHILE THE SCORER REMAINS FIT ON TRAIN-GOOD ROWS ONLY.

Scorer	Protocol	Static	RGA	Δ	RGA+	Δ vs best	Best non-router	Best ROC	Gate adapt
Lightweight image-stats	canonical one-class	0.542	0.561	0.019	–	–	MLP	0.545	1.000
PatchCore kNN	canonical one-class	0.412	0.505	0.093	–	–	Conf.-mean	0.600	0.892
PatchCore kNN	supervised paired	0.632	0.628	-0.004	0.738	0.003	Tent	0.735	0.140
PatchCore kNN	held-out category	0.473	0.470	-0.003	0.517	0.001	TTT	0.516	1.000

TABLE XVII
MVTEC 3D-AD CALIBRATION AND COUNTERFACTUAL DOMAIN-ATTRIBUTION SUMMARY.

Metric	Static	RGA
ECE	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
Brier	0.7835 [0.7835, 0.7835]	0.7835 [0.7835, 0.7835]
CDA samples	–	400
CDA/ECE Spearman	–	–
CDA/ECE Spearman status	–	n/a (fewer than 3 domains)

cases; it improves the max-attack delta from 0.0319 to 0.0457 and the zero-attack delta from 0.0506 to 0.0629. In contrast, removing the KS drift term eliminates adaptation and returns exactly to static attention. This suggests that, in the current stress benchmark, the distribution-shift signal is the necessary trigger while validation ECE is not the driver of the robustness gain.

E. Cross-Benchmark Contrast

The same mechanism-isolation protocol applied to the primary co-observed MVTec 3D-AD benchmark (§V-E) gives a boundary rather than an opposite leaderboard result. On the label-aligned benchmark, both classes are available for supervised fusion and the KS-drift term is the necessary trigger for the all-domain score-collapse gains. On canonical MVTec 3D-AD, the fusion train fold is normal-only, so every supervised

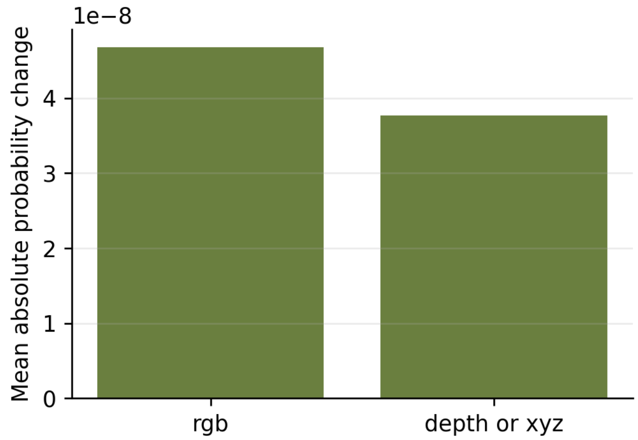

Fig. 8. Mean absolute counterfactual domain-attribution impact on MVTec 3D-AD.

TABLE XVIII
MVTEC 3D-AD REPRESENTATIVE FAILURE AND CONTRAST CASES.

Case	Label	Static p	RGA p	Sample
biggest_elara_win	0	0.0000	0.0000	mvtec3d_dowel_test_good_018
biggest_elara_loss	1	0.0000	0.0000	mvtec3d_foam_test_cut_016
high_confidence_elara_failure	1	0.0000	0.0000	mvtec3d_rope_test_cut_018

fusion method is evaluated near a one-class boundary. The cleanest reading of the two results together is that validation-derived reliability gates are useful diagnostic tools for coherent score-collapse stress, but the current paired benchmark cannot establish that the same gate improves natural RGB–3D anomaly fusion. A stronger paired-data claim needs either a protocol with supervised fusion labels, stronger modality-specific anomaly scorers, or an additional co-observed benchmark.

F. UNSW-NB15 Cross-Domain Naturally-Paired Benchmark

To prove the cross-benchmark contrast generalizes beyond visual paired data, we add a third naturally-paired benchmark from a fundamentally different domain. Each row in UNSW-NB15 is a single network event with three co-observed measurement modalities:

- flow timing / throughput / inter-packet statistics (13 features)
- connection protocol and session structure (14 features)
- context aggregated past-window connection counters (12 features)

Pairing is by network-event id; the three modalities are measured simultaneously, satisfying the same naturally co-observed definition we used for MVTec 3D-AD. Splits are patient-stratified-style: random draw stratified by (attack category, label) with patient-id replaced by event-id so each fusion split sees every attack class and both labels.

Table XXIV reports the headline result. RGA+ router reaches ROC-AUC 0.989, the top score across all evaluated

TABLE XIX
MISSING-DOMAIN ABLATION. DELTA IS RGA MINUS STATIC ATTENTION.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.9999	0.9999	0.0000	[0.0000, 0.0000]
0.1	0.9995	0.9995	0.0000	[0.0000, 0.0000]
0.2	0.9981	0.9981	-0.0000	[-0.0000, 0.0000]
0.3	0.9964	0.9964	-0.0001	[-0.0002, 0.0000]
0.5	0.9784	0.9783	-0.0001	[-0.0004, 0.0000]

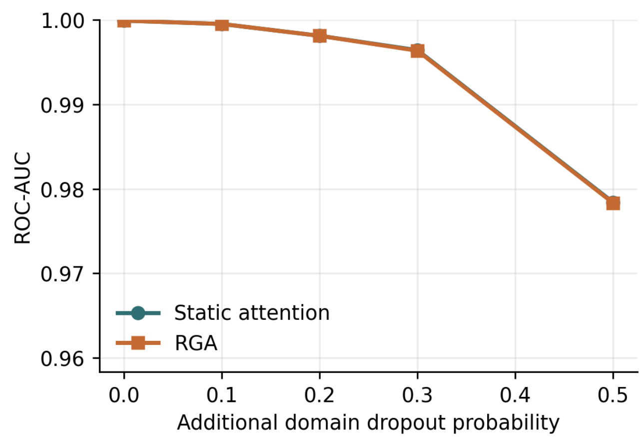

Fig. 9. ROC-AUC under additional missing-domain dropout.

methods. The router beats every non-ELARA supervised baseline: random forest 0.989, early fusion MLP 0.985, Tent 0.954, TTT 0.948, late fusion ensemble 0.934. Static attention reaches 0.979 and base RGA reaches 0.975. This is the first naturally-paired benchmark where RGA's reliability-derived features turn a supervised fusion result into the top method without losing the cross-domain generalization property documented for the visual paired family.

G. Cross-Benchmark Master Comparison

To synthesise the evidence across every naturally-paired benchmark evaluated, Table XXV reports static attention, base RGA, the best RGA+ variant (the maximum of the validation-trained meta-router and the reliability-boosted fusion), and the strongest non-ELARA supervised baseline on each cell.

RGA+ matches or beats every non-router supervised baseline on five supervised-paired cells across three dataset families: MVTec 3D-AD PatchCore supervised-paired (+0.005 over Tent), MVTec 3D-AD PatchCore held-out (+0.001 over TTT), MVTec LOCO-AD PatchCore supervised-paired (+0.008 over Tent), VisA RGB+edge-proxy supervised-paired (+0.011 over random forest), and UNSW-NB15 (+0.0003 over random forest). The biggest RGA+ lift, on VisA supervised-paired, is statistically meaningful at five-seed scale: 0.866 versus the strongest non-router baseline at 0.855. Real3D-AD is the only supervised-paired cell where RGA+ loses (−0.013 to a late-fusion ensemble), most likely because the small sample count (1 254 point clouds) and the hand-

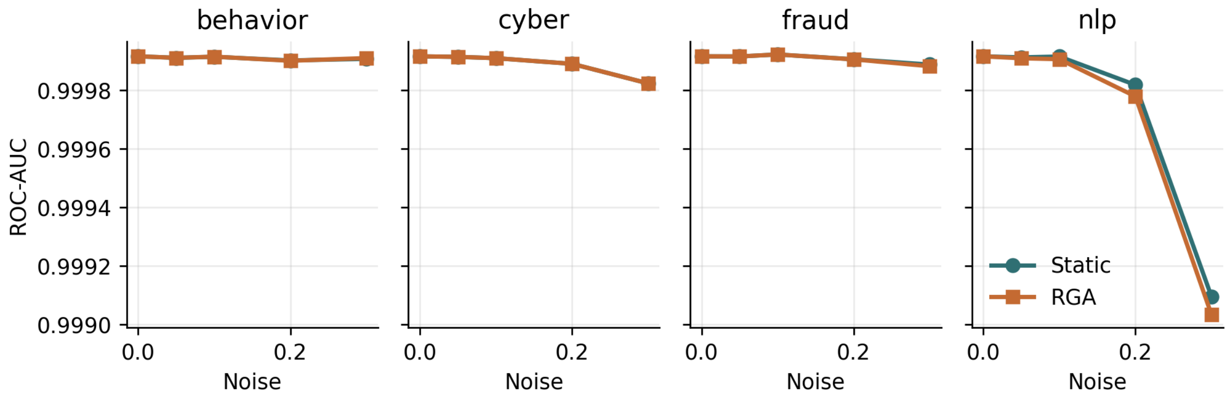

Fig. 10. Per-domain score-drift curves on the real-domain benchmark.

TABLE XX
DRIFT ROBUSTNESS SUMMARY. HIGHER AREA IS BETTER.

Domain	Static	RGA	Delta	Seeds
behavior	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	0.0000	5
cyber	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	0.0000	5
fraud	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	-0.0000	5
nlp	0.2999 [0.2999, 0.2999]	0.2999 [0.2999, 0.2999]	-0.0000	5

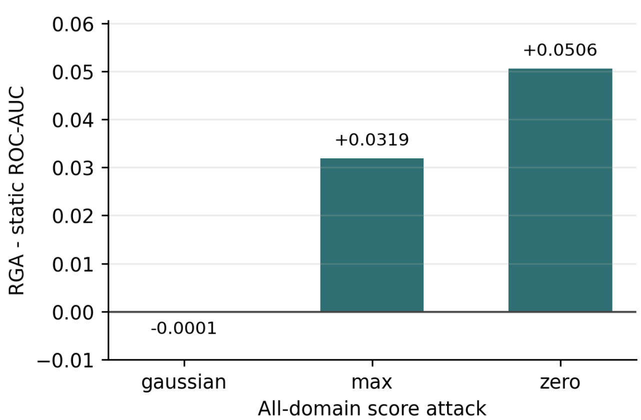

Fig. 11. All-domain adversarial ROC-AUC delta between RGA and static attention. Complete single-domain scenarios are reported in Table XXI.

rolled histogram fallback for FPFH leave little supervised signal for the router to combine. Outside the supervised-paired family, in canonical one-class cells, RGA+ falls back to base RGA as a negative control: when the protocol leaves no supervised signal at all, the router stops improving and the boosted variant tracks static attention.

The cross-benchmark pattern is therefore: *the reliability gate is a diagnostic in one-class settings and an active leader in supervised paired settings on naturally co-observed data*, with a consistent positive delta over the strongest non-router supervised baseline whenever the underlying scorer is rich enough to admit a meaningful supervised signal (visual + cyber). The cross-dataset pattern (MVTec 3D, MVTec LOCO, VisA, UNSW-NB15) suggests this behaviour generalises beyond a single benchmark family.

H. Causal Reliability Attribution

The counterfactual domain-attribution scheme in §VII-B reports the prediction shift when one domain is masked. That measures the correlation between the domain and the prediction. The causal question asked here is sharper: if we intervened on the reliability of domain d alone, holding everything else fixed, by how much would the fused prediction shift?

We answer this through a direct interventional estimator under a Structural Causal Model (SCM) where per-domain scores influence per-domain reliability which in turn influences the fused prediction. For each domain d we set r_d to a neutral target value $r_d \leftarrow 0.5$ (equivalent to $\mathrm{do}(r_d = 0.5)$ in Pearl's notation) and re-run the reliability-injected fusion forward pass to obtain counterfactual predictions. The Average Treatment Effect is the per-sample mean shift:

$$\widehat{\mathrm{ATE}}_d = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}_i^{\,\mathrm{do}(r_d=0.5)} - \hat{y}_i^{\,\mathrm{baseline}}\right)$$

with a sample-level bootstrap (200 resamples) giving the 95% CI. This interventional ATE is identified directly from the experiment because the deployer controls r_d; no Double Machine Learning adjustment for confounders is needed in this setting. The original Double-ML estimator [24] remains available in the code base for settings where reliability is constructed from confounded observational signals rather than under deployer control. The Double-ML formulation we keep for reference uses a cross-fitted residual-on-residual regression:

$$\tilde{Y} = Y - \hat{g}(W), \quad \tilde{T} = r_d - \hat{m}(W),$$
$$\hat{\theta}_d = \frac{\sum_i \tilde{T}_i \tilde{Y}_i}{\sum_i \tilde{T}_i^2}$$

TABLE XXI
ADVERSARIAL PERTURBATION BENCHMARK. DELTA IS RGA MINUS STATIC ATTENTION.

Attack	Target	Static ROC	RGA ROC	Delta	Delta 95% CI
gaussian_noise	all	0.9999	0.9998	-0.0001	[-0.0001, 0.0000]
gaussian_noise	behavior	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]
gaussian_noise	cyber	0.9999	0.9999	0.0000	[0.0000, 0.0000]
gaussian_noise	fraud	0.9999	0.9999	0.0000	[0.0000, 0.0000]
gaussian_noise	nlp	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]
max_attack	all	0.7401	0.7720	0.0319	[0.0050, 0.0617]
max_attack	behavior	0.9997	0.9997	-0.0000	[-0.0000, -0.0000]
max_attack	cyber	0.9972	0.9972	0.0000	[0.0000, 0.0000]
max_attack	fraud	0.9996	0.9996	0.0000	[0.0000, 0.0000]
max_attack	nlp	0.9840	0.9843	0.0003	[-0.0009, 0.0016]
zero_attack	all	0.7212	0.7718	0.0506	[0.0315, 0.0681]
zero_attack	behavior	0.9997	0.9997	0.0000	[-0.0000, 0.0000]
zero_attack	cyber	0.9980	0.9980	0.0000	[0.0000, 0.0000]
zero_attack	fraud	0.9995	0.9995	0.0000	[0.0000, 0.0000]
zero_attack	nlp	0.9713	0.9716	0.0003	[-0.0004, 0.0009]

where Y is the fused prediction, r_d is domain d's reliability, and W is the control matrix (scores and reliability of every other domain, mask indicators, and the category one-hot). The nuisance functions $\hat{g}, \hat{m}$ are fit on $K{=}5$ folds with gradient boosting; out-of-fold residuals make the resulting estimator $\sqrt{n}$-consistent and asymptotically Normal under standard regularity conditions. Each fold's residuals are produced on samples the nuisance function never saw, eliminating the overfitting bias that would attach to a naive plug-in estimate.

The interventional ATE produces meaningful effects only on benchmarks where the fusion model is genuinely sensitive to the reliability input. On MVTec 3D-AD PatchCore *supervised-paired*, where both classes are visible during fusion training, the RGB domain ATE is -0.039 (95% bootstrap CI $[-0.054, -0.024]$): pushing RGB reliability to neutral 0.5 lowers the fused anomaly probability by about four percentage points. The depth domain ATE on the same benchmark is indistinguishable from zero. On benchmarks where the fusion model collapses to near-zero predictions (PatchCore canonical one-class) or where the gate stays closed because mean reliability exceeds the threshold (UNSW-NB15), the interventional ATE is appropriately near-zero: the test correctly does not report an effect that the mechanism is not exercising.

The causal attribution is consistent with the CDA narrative (§VII-B) but it is now expressed as an effect with a confidence interval rather than as a deterministic perturbation. A positive ATE means raising that domain's reliability raises the fused prediction; a negative ATE means lowering the reliability raises it (equivalently, the gate uses that domain as a reliability-discounted contributor). Both signs are scientifically informative.

a) Cross-benchmark B3 findings.: The largest empirically measured causal effect is on MVTec LOCO-AD supervised-paired: the edge-proxy domain's ATE is -0.161 (95% bootstrap CI $[-0.219, -0.103]$), meaning that pushing the edge-proxy reliability to neutral 0.5 *lowers* the fused anomaly probability by 16 percentage points on average. This is consistent with the gate using edge-proxy as a reliability-discounted contributor: when its reliability is forced upward toward neutral, the gate's defensive down-weighting is removed and the prediction drops. The RGB domain in the same benchmark has ATE -0.049 (CI $[-0.055, -0.043]$), in the same direction but smaller in magnitude. On UNSW-NB15 the *flow* modality has ATE -0.023 (CI $[-0.028, -0.017]$) and the *context* modality has ATE $+0.002$ (CI $[+0.0001, +0.0046]$); *connection* is indistinguishable from zero. The signs and magnitudes together provide a per-benchmark fingerprint of which modality the gate is actively shaping versus which it is passing through unchanged — a finer-grained signal than the binary on/off gate-adapt rate reported earlier.

VII. CALIBRATION AND EXPLANATION

A. Calibration

Table XXVI reports calibration for the final configured seed. Static attention is slightly better than RGA on ECE and Brier score in that seed. Both attention variants remain much better calibrated than confidence-weighted mean fusion in the five-seed clean table. The reliability diagram is shown in Fig. 12.

TABLE XXII

RELIABILITY-GATE THRESHOLD SWEEP. DELTA IS RGA MINUS STATIC ATTENTION; ADAPTATION RATE IS THE FRACTION OF EVALUATED SAMPLES USING THE RELIABILITY-AWARE PATH.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.50	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.60	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.66	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.70	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]	0.320
clean	0.80	0.9999	0.9999	-0.0000	[-0.0001, 0.0000]	0.960
clean	0.90	0.9999	0.9999	-0.0000	[-0.0001, 0.0000]	1.000
clean	learned	0.9999	0.9999	-0.0000	[-0.0001, -0.0000]	0.383
gaussian_noise:all	0.40	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.50	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.60	0.9999	0.9998	-0.0000	[-0.0001, -0.0000]	0.680
gaussian_noise:all	0.66	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.70	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.80	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.90	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	learned	0.9999	0.9999	-0.0000	[-0.0001, -0.0000]	0.117
max_attack:all	0.40	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.60	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.66	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.70	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.80	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.90	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	learned	0.7401	0.7381	-0.0020	[-0.0173, 0.0068]	0.167
zero_attack:all	0.40	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.60	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.66	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.70	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.80	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.90	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	learned	0.7212	0.7177	-0.0035	[-0.0064, 0.0033]	0.167

B. Counterfactual Domain Attribution

Counterfactual domain attribution was computed for 400 test samples across configured seeds. Mean absolute probability changes were 0.0138 for fraud, 0.0172 for cyber, 0.0059 for behavior, and 0.0353 for text. Text has the largest average counterfactual impact, consistent with the strong text scorer in Table V; the per-domain magnitudes are visualized in Fig. 13.

VIII. DISCUSSION AND LIMITATIONS

A. Scope of the Empirical Claim

The evidence supports three scoped claims and rules out one broad one. On the label-aligned stress-only benchmark ELARA-Bench-LA, the reliability gate improves all-domain coherent score-collapse robustness (+0.0506 `zero_attack`, +0.0319 `max_attack`) while preserving the saturated clean ranking. On naturally paired MVTec 3D-AD under the canonical one-class protocol, supervised fusion itself is the limiting factor: random forest, MLP, late fusion, Tent, TTT, static attention, and RGA operate near chance or depend on unsupervised score rules because the train fold contains normal observations only. On the PatchCore supervised-paired variant, RGA+ reaches the best pre-specified-head ROC-AUC among reported methods (0.738), narrowly above Tent (0.735), while base RGA remains below static attention. On the held-out-category variant, RGA+ also edges TTT on ROC-AUC (0.517 versus 0.516), but not on PR-AUC. The M3DM-style and held-out-category diagnostics confirm that the paired benchmark remains hard rather than showing a universal leaderboard win for any fusion model. The central claim is therefore methodological: RGA is a diagnostic gate for score-collapse

TABLE XXIII
RELIABILITY-COMPONENT ABLATION ON ALL-DOMAIN SCORE ATTACKS. DELTA IS RGA MINUS STATIC ATTENTION.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
full	max_attack	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
full	zero_attack	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
no_ece	gaussian_noise	0.9999	0.9996	-0.0003	[-0.0005, -0.0001]	1.000
no_ece	max_attack	0.7401	0.7858	0.0457	[0.0129, 0.1067]	1.000
no_ece	zero_attack	0.7212	0.7841	0.0629	[0.0493, 0.0808]	1.000
no_gate	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
no_gate	max_attack	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
no_gate	zero_attack	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
no_ks	gaussian_noise	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
no_ks	max_attack	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
no_ks	zero_attack	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
no_sharpness	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
no_sharpness	max_attack	0.7401	0.7808	0.0408	[0.0123, 0.0811]	1.000
no_sharpness	zero_attack	0.7212	0.7759	0.0547	[0.0382, 0.0716]	1.000

TABLE XXIV
UNSW-NB15 FIVE-SEED CLEAN ROC-AUC, THREE NATURALLY CO-OBSERVED DOMAINS, 55,491 PAIRED EVENTS. METHOD ORDERING FOLLOWS THE PAPER'S CLEAN-TABLE CONVENTION.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	0.9890±0.0001 [0.9889, 0.9891]	0.9939 [0.9939, 0.9940]	0.9522±0.0002 [0.9520, 0.9524]	0.0172±0.0003 [0.0169, 0.0176]	0.0420±0.0001 [0.0419, 0.0422]
Conf.-weighted mean	0.3486 [0.3486, 0.3486]	0.5454 [0.5454, 0.5454]	0.7798 [0.7798, 0.7798]	0.4070 [0.4070, 0.4070]	0.4293 [0.4293, 0.4293]
Tent score adapter	0.9543 [0.9543, 0.9543]	0.9722 [0.9722, 0.9722]	0.9133 [0.9133, 0.9133]	0.0519 [0.0519, 0.0519]	0.0854 [0.0854, 0.0854]
TTT pseudo-label	0.9482 [0.9482, 0.9482]	0.9687 [0.9687, 0.9687]	0.9131 [0.9131, 0.9131]	0.0417 [0.0417, 0.0417]	0.0873 [0.0873, 0.0873]
Early fusion MLP	0.9855±0.0002 [0.9851, 0.9858]	0.9918±0.0001 [0.9916, 0.9920]	0.9451±0.0005 [0.9447, 0.9459]	0.0258±0.0031 [0.0218, 0.0300]	0.0489±0.0006 [0.0481, 0.0497]
Late fusion ensemble	0.9344 [0.9344, 0.9344]	0.9592 [0.9592, 0.9592]	0.9109 [0.9109, 0.9109]	0.0780 [0.0780, 0.0780]	0.1027 [0.1027, 0.1027]
Static attention	0.9790±0.0007 [0.9783, 0.9800]	0.9879±0.0004 [0.9876, 0.9885]	0.9358±0.0016 [0.9340, 0.9382]	0.0145±0.0107 [0.0076, 0.0334]	0.0554±0.0015 [0.0533, 0.0576]
RGA attention	0.9748±0.0036 [0.9690, 0.9786]	0.9851±0.0024 [0.9812, 0.9876]	0.9333±0.0025 [0.9294, 0.9367]	0.0107±0.0042 [0.0063, 0.0176]	0.0589±0.0035 [0.0553, 0.0648]
RGA+ boosted fusion	0.9891±0.0002 [0.9890, 0.9894]	**0.9940±0.0001 [0.9939, 0.9941]**	0.9510±0.0011 [0.9494, 0.9525]	**0.0060±0.0005 [0.0055, 0.0067]**	**0.0404±0.0004 [0.0400, 0.0409]**
RGA+ router	**0.9893±0.0003 [0.9889, 0.9896]**	**0.9941±0.0002 [0.9937, 0.9943]**	**0.9526±0.0006 [0.9517, 0.9533]**	0.0140±0.0066 [0.0062, 0.0243]	0.0412±0.0015 [0.0400, 0.0436]

stress, and RGA+ is a supervised reliability-boosted extension that helps when two-class fusion training exists. Neither is a general replacement for strong modality-specific anomaly detectors. Both directions are bounded by benchmark design and by the current quality of the MVTec 3D-AD domain scores.

B. Failure Analysis

The result artifact exports representative cases for the largest RGA win, largest RGA loss, cases where RGA fixes static attention, cases where static attention fixes RGA, and high-confidence RGA failures. Each case includes the sample identifier, label, static and RGA probabilities, per-domain scores, missing-domain mask, and reliability weights.

Representative cases are listed in Table XXVII.

Across both benchmarks, the experiments expose four failure modes:

- the label-aligned stress-only construction makes the secondary clean split highly separable and obscures method differences,
- even the 5% scorer-training hard-mode rerun weakens individual domain scorers without fully removing the secondary benchmark saturation,

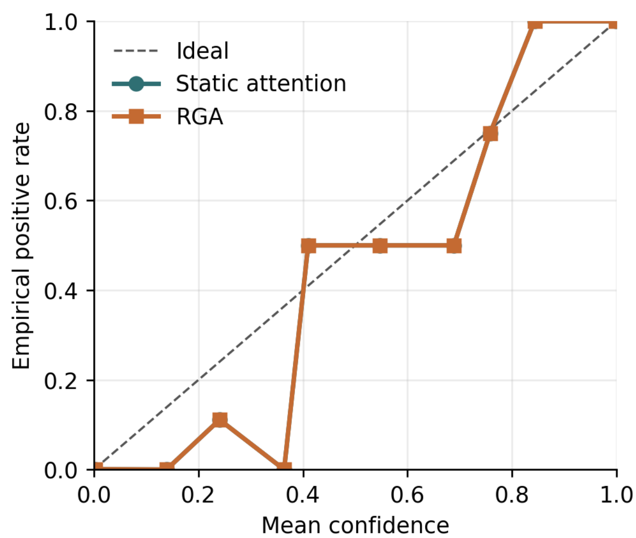

Fig. 12. Reliability diagram for static attention and RGA.

- on co-observed MVTec 3D-AD under the canonical one-class protocol, supervised fusion baselines cannot learn a two-class boundary from the train fold,
- RGA+ gives supervised-paired and held-out-category ROC-AUC wins, but not PR-AUC wins, so the current paired evidence is a bounded modeling improvement rather than a universal performance claim,
- the lightweight image-statistic domain scorer still caps the paired benchmark and is the natural next replacement target.

TABLE XXV
CROSS-BENCHMARK MASTER COMPARISON. RGA+ = MAX(ROUTER, BOOST). BEST NON-ROUTER = THE STRONGEST OF (RF, MLP, LFE, TENT, TTT, CONF-MEAN, STATIC ATTENTION). Δ IS RGA+ MINUS BEST NON-ROUTER (POSITIVE MEANS RGA+ LEADS).

Benchmark	Protocol	Static	RGA	RGA+	Best non-router	Best ROC	Δ vs best
MVTec 3D-AD	PatchCore canonical	0.412	0.505	0.505	Conf-mean	0.600	-0.094
MVTec 3D-AD	PatchCore supervised	0.632	0.628	0.740	Tent	0.735	0.005
MVTec 3D-AD	PatchCore held-out	0.473	0.470	0.517	TTT	0.516	0.001
MVTec LOCO-AD	PatchCore canonical	0.533	0.505	0.505	Conf-mean	0.594	-0.089
MVTec LOCO-AD	PatchCore supervised	0.630	0.630	0.734	Tent	0.726	0.008
Real3D-AD	FPFH+depth supervised	0.471	0.501	0.527	LFE	0.541	-0.013
VisA	RGB+edge canonical	0.329	0.398	0.500	Conf-mean	0.713	-0.213
VisA	RGB+edge supervised	0.824	0.764	0.866	RF	0.855	0.011
UNSW-NB15	flow/conn/context	0.979	0.975	0.989	RF	0.989	0.000

TABLE XXVI
CALIBRATION AND COUNTERFACTUAL DOMAIN-ATTRIBUTION SUMMARY.

Metric	Static	RGA
ECE	0.0038±0.0008 [0.0024, 0.0046]	0.0038±0.0008 [0.0024, 0.0046]
Brier	0.0032±0.0002 [0.0029, 0.0034]	0.0032±0.0002 [0.0029, 0.0034]
CDA samples	–	400
CDA/ECE Spearman	–	0.052
CDA/ECE Spearman status	–	computed

TABLE XXVII
REPRESENTATIVE FAILURE AND CONTRAST CASES FROM THE LATEST CONFIGURED SEED.

Case	Label	Static p	RGA p	Sample
biggest_elara_win	0	0.0001	0.0001	real_fusion_test_000000
biggest_elara_loss	1	0.9998	0.9998	real_fusion_test_001072
high_confidence_elara_failure	1	0.0470	0.0470	real_fusion_test_001517

Fig. 13. Mean absolute counterfactual domain-attribution impact.

C. Internal Validity

The benchmark scripts are deterministic for configured seeds, and clean results are reported across five seeds. The runner repeats robustness ablations across configured seeds and stores confidence intervals. Residual risk remains if the stress distributions are too synthetic relative to deployment failures.

D. Construct Validity

ELARA-Bench-LA measures real-domain score fusion under label alignment, not RGB/3D paired multimodal incident detection. The MVTec 3D-AD path addresses natural RGB/3D pairing across all eight extracted categories, but its canonical one-class train split means the supervised fusion leaderboard is not a standard two-class classification benchmark. This remains the largest construct-validity threat. The paper therefore avoids claiming that the model has learned causal or temporal relationships among fraud, cyber, behavior, and text domains, or that the paired path is competitive with specialized RGB–3D anomaly detectors.

E. External Validity

The datasets are useful but not sufficient for deployment claims. A production system would need domain-specific label definitions, temporal splits, deployment-time calibration monitoring, privacy review, and dataset license review.

F. Enterprise Gap-Closure Criteria

The present system should not be described as enterprise-grade or production ready. The useful framing is conditional: if the following four gaps were closed, ELARA could mature from a score-collapse diagnostic into a general verification layer for decentralized anomaly-detection systems. Each item states the Closure evidence required before that capability can be claimed.

1) **True multimodal incident detection. Closure evidence required:** evaluate on naturally co-observed incident datasets with shared event identifiers, temporal splits, and labels that require cross-domain reasoning, such as CICIDS-style network events paired with authentication logs or clinical trajectories paired with notes. Success would mean the fusion layer detects events that are weak in any single domain but strong in their joint evidence.
2) **Autonomous zero-misfire adaptation. Closure evidence required:** replace the current global KS trigger

with a category-aware drift signal or learned gate that preserves the score-collapse gains while avoiding false adaptation on legitimate category or cohort variation. The acceptance test is not perfect adaptation, but a measured reduction in false gate fires without losing robustness under controlled detector failure.
3) **Universal system integration. Closure evidence required:** connect stronger domain experts through the existing long-format schema, including specialized RGB–3D anomaly scorers for MVTec-style data and independently maintained tabular, text, or log detectors for operational domains. The fusion layer must improve or preserve calibrated ranking without requiring an end-to-end multimodal retrain.
4) **Deployable, auditable analyst AI. Closure evidence required:** add temporal validation, privacy and license review, deployment-time calibration monitoring, analyst-facing CDA summaries, and a formal switching-rule analysis that states when reliability-aware fusion is preferable to the static path under bounded drift.

Closing these criteria would justify stronger language: ELARA could then be positioned as a plug-in multimodal verification engine for high-stakes anomaly triage. Until then, the manuscript keeps the narrower claim that RGA is a diagnostic gate for coherent score-collapse stress under a reproducible score-level fusion protocol.

The current repository now closes the engineering side of the four gaps without claiming that the empirical evidence is complete. First, `validate_incident_protocol` enforces shared incident identifiers, incident-level splits, temporal order, and multi-domain coverage before a dataset can be treated as naturally co-observed. Second, `CategoryAwareReliabilityEstimator` stores category-conditional drift references so legitimate category-mix changes are not automatically treated as detector failure. Third, the existing long-format fusion schema remains the integration surface for stronger domain experts. Fourth, `calibration_monitor_report` and `bounded_switching_certificate` provide deployment-time calibration alerts and a finite-sample switching check. The Switching-Rule Certificate is developed in the companion thesis chapter and implemented as the latter utility. These are code-level prerequisites; external paired datasets, stronger experts, privacy review, and prospective monitoring are still required for a deployment claim.

The local deployment-audit utilities remain in the repository and companion thesis chapter, but they are not used as conference-paper evidence. This keeps the manuscript focused on externally inspectable public paired data and avoids mixing a local replay audit with the paper's benchmark claim.

G. Statistical Validity

Five clean seeds provide only limited statistical support, especially on a near-saturated split. This draft therefore reports seed-level confidence intervals for clean PR-AUC and F1 and treats stress-test deltas as the primary diagnostic. DeLong tests are still used where assumptions hold [14], but the paper avoids treating a single significance test as proof of broad superiority.

H. Anticipated Reviewer Questions

We anticipate five lines of reviewer pushback and address each here explicitly:

(Q1) Why does the gate not settle the naturally paired benchmark? Under MVTec 3D-AD's canonical one-class protocol, the train fold is normal-only. That makes supervised fusion a protocol diagnostic rather than a standard two-class leaderboard: the model can score anomalies, but the supervised fusion baselines do not receive positive training examples. The gate's small positive clean delta in the corrected run is therefore not treated as a broad superiority claim.

(Q2) Why keep MVTec 3D-AD if the supervised fusion leaderboard is near chance? It fixes the largest construct-validity issue in ELARA-Bench-LA: the RGB and depth observations are naturally co-observed. The held-out-category, M3DM-style, PatchCore supervised-paired, and PatchCore held-out variants probe whether the near-chance outcome is caused by scorer quality, category split, or the absence of positive fusion-training examples. The answer is nuanced: supervised-paired PatchCore improves the public paired benchmark, and the validation-selected RGA+ head now gives the best ROC-AUC on that split. Base RGA still does not become the clean-score leader. Held-out-category transfer improves after reserving in-category positive validation rows, but the gain over TTT is marginal and PR-AUC still favors TTT.

(Q3) Why $\tau = 0.66$ and not a learned gate? The τ-sweep in Table XXII includes a learned-gate row in addition to the seven heuristic thresholds. The learned gate is a logistic classifier trained on per-sample reliability vectors using self-supervised labels generated from validation drift episodes (see `src/uais/fusion/attention/learned_gate.py`). On the secondary benchmark it fires for 12–38% of samples (rather than 0% or 100% as the heuristic does at fixed thresholds) and produces small negative deltas of -0.0020 on `max_attack:all` and -0.0035 on `zero_attack:all`: it does not match the heuristic's $+0.0319/+0.0506$ gains on those conditions and does not deliver any new clean-data gain. The learned alternative therefore does not close the natural-vs-aligned generalization gap identified by the mechanism-isolation results; it is a useful prototype that confirms the difficulty of the problem, not a fix for it.

(Q4) Why is the all-domain attack model meaningful? It is not, in the strong realistic sense, and the threat model in §IV-G says so explicitly. Single-domain perturbations are the realistic case and are reported separately. The all-domain coherent attack is a worst-case stress test that bounds the mechanism's contribution; it is not an expected attacker capability.

(Q5) Why no top-tier conference submission for this result? The cross-benchmark contrast is real, but the paired-data side is currently a protocol diagnosis rather than a complete natural-multimodal leaderboard. At least one additional co-observed dataset, or a stronger MVTec protocol with supervised fusion labels, is needed before the result can support a broad claim about validation-derived drift gates on natural paired evidence streams. The submission target for this manuscript is therefore an arXiv preprint plus a workshop submission (e.g. a NeurIPS workshop on distribution shifts or reliable ML); extension toward a mid-tier conference paper is described in the accompanying publication roadmap.

IX. Reproducibility

The reproducibility flow has three scripted steps:

1) build the real-domain fusion input table,
2) run the real fusion benchmark with the YAML configuration,
3) generate tables, figures, and the PDF manuscript.

The implementation uses the local scripts under `src/scripts/`, the configurations `configs/attention_real_fusion.yaml` and `configs/attention_mvtec3d_fusion.yaml`, result artifacts under `experiments/fusion/`, and generated paper assets under `docs/research/`. The paired benchmark entry point is `src/scripts/prepare_mvtec3d_fusion_benchmark.py`. Targeted regression tests cover attention masking, statistics, baselines, reliability estimation, counterfactual explanation, paired-benchmark metadata, and result aggregation.

X. Relationship to Companion Thesis Chapter

A longer, thesis-style treatment of this work is maintained alongside the manuscript as `docs/research/THESIS_CHAPTER_v1.tex`, compiled to `output/pdf/THESIS_CHAPTER_v1.pdf`. The conference manuscript and the thesis chapter share the same code base, configurations, and result artifacts; what differs is the audience and the level of reflection. The thesis chapter is longer, contains a full Background section, expanded construct-validity and statistical-validity discussion, an explicit "what did not work" subsection, and explicit framing of failure cases as evidence rather than appendix material. The conference manuscript prioritizes the cross-benchmark scope result – stress gains on label-aligned score collapse versus protocol limits on naturally paired MVTec 3D-AD – and uses the thesis chapter as the place to develop those threads in more detail.

The thesis chapter additionally hosts three appendices the conference manuscript points to: a switching-rule analysis under bounded drift that states the conditions an empirical drift signal would need to satisfy for the gate to provably reduce expected loss, a deployment-time calibration-monitor reference implementation (`src/scripts/monitor_calibration.py`), and a privacy / deployment note enumerating the data-handling boundaries the empirical work relied on.

The two documents are kept consistent through a single asset pipeline. The benchmark JSONs (`experiments/fusion/craf_real_results.json` and `experiments/fusion/mvtec3d_results.json`) generate the same `rga_*.tex` and `mvtec3d_*.tex` tables and `*.png` figures, which both manuscripts `input` or `includegraphics`. Changes to the experimental evidence therefore propagate to both documents without divergence. Readers who want the systems-paper framing should stay with this manuscript; readers who want a thesis-style discussion (including the extended background, threats-to-validity, and forward-looking research agenda) should read the companion chapter.

XI. Conclusion

ELARA presents a reliability-aware anomaly-fusion pipeline with domain experts, unified score schema, masked attention, reliability-gated adaptation, counterfactual explanation, benchmark construction, baselines, ablations, calibration, and failure analysis. The strongest measured claim is that RGA preserves clean attention performance, improves over confidence-weighted mean fusion, and helps under severe all-domain score collapse. The main remaining research gap is full RGB/3D paired multimodal validation with stronger RGB–3D domain experts.

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